# Interval search experiments

Built 2026-10-17 03:42 PT. Experiment design by Roland Gao; runs and analysis by Claude. Code: research/cooldown_search_probe.py (runs), research/cooldown_study.py (analysis), research/make_cooldown_study.py (this PDF). Logs: logs/research_cooldown_search_probe_*, logs/research_hparam_importance_*. Batch size 2000 throughout.

## Experiment 1. Cooldown study: which 40-step lookahead ranks interval candidates like the final result?

### *Question*

An interval search picks each interval's LR by training the interval with a candidate LR, then a short lookahead (the cooldown), and scoring the result. Which cooldown ranks the candidates the way the final accuracy does? Compared, all with a 40-step lookahead (one interval at batch size 2000): no lookahead; linear decay to 0 (conv only, or all four groups); one searched constant cooldown, either for conv only with the other groups on their ground-truth schedules, or shared by all four groups; and four searched constant cooldowns (one per group).

### *Setup*

| Item | Value |
|---|---|
| Model / data | CIFAR-10 airbench, batch size 2000, 200 steps, baseline_bs2000 (conv Muon2 0.22 -> 0; head, norm_bias, whiten_bias SGD on their baseline schedules) |
| Intervals | 4 intervals of 40 steps: steps 0-40, 40-80, 80-120, 120-160 (the last interval has no room for a 40-step lookahead) |
| Candidates | 7 constant conv LRs for the interval: the 0.6^k grid point nearest the baseline's mean conv LR in that interval, and 3 grid steps either side; other groups on the baseline |
| Ground truth | Final test TTA when training continues on the baseline schedule after the interval |
| No lookahead | Test TTA right after the interval |
| Linear to 0 | Conv only: conv decays linearly from the candidate to 0 over 40 steps, the other groups continue their schedules. All groups: every group decays linearly to 0 from its current value |
| 1 searched cooldown: conv only | Conv constant at m x candidate for 40 steps; the other groups follow their baseline (ground-truth) schedules; m chosen from {1, 0.6, 0.36, 0.22, 0.13, 0.078, 0.047, 0} by the best score on that seed |
| 1 searched cooldown: shared | Constant LRs for 40 steps: conv at m x candidate, each other group at m x its mean LR over the interval; one m for all groups, chosen as above |
| 4 searched cooldowns | As above but one m per group; coordinate search from the best shared m, one group at a time over the same 8 values, until no group changes (at most 3 passes) |
| Seed | 1 training seed (0); every method uses the same prefix and data order |
| Score | Test-set TTA accuracy (%) |

### *Results*

Picked LR = the candidate with the highest score. Regret = ground truth at its best candidate minus ground truth at the picked one (pp). One seed: single-run noise is about 0.13 pp, so differences of that size between candidates are not resolved.

### *Picked LR (regret, pp)*

| Method | Interval 1 | Interval 2 | Interval 3 | Interval 4 |
|---|---|---|---|---|
| Ground truth (final TTA) | 0.078 (0.00) | 0.13 (0.00) | 0.078 (0.00) | 0.078 (0.00) |
| No lookahead | 0.078 (0.00) | 0.028 (0.34) | 0.028 (0.01) | 0.028 (0.29) |
| Linear to 0: conv only | 0.13 (0.05) | 0.078 (0.26) | 0.078 (0.00) | 0.028 (0.29) |
| Linear to 0: all four groups | 0.13 (0.05) | 0.078 (0.26) | 0.047 (0.21) | 0.028 (0.29) |
| 1 searched cooldown: conv only, others on ground truth | 0.13 (0.05) | 0.078 (0.26) | 0.047 (0.21) | 0.047 (0.21) |
| 1 searched cooldown: shared by all four groups | 0.13 (0.05) | 0.078 (0.26) | 0.047 (0.21) | 0.047 (0.21) |
| 4 searched cooldowns (one per group) | 0.13 (0.05) | 0.078 (0.26) | 0.047 (0.21) | 0.047 (0.21) |

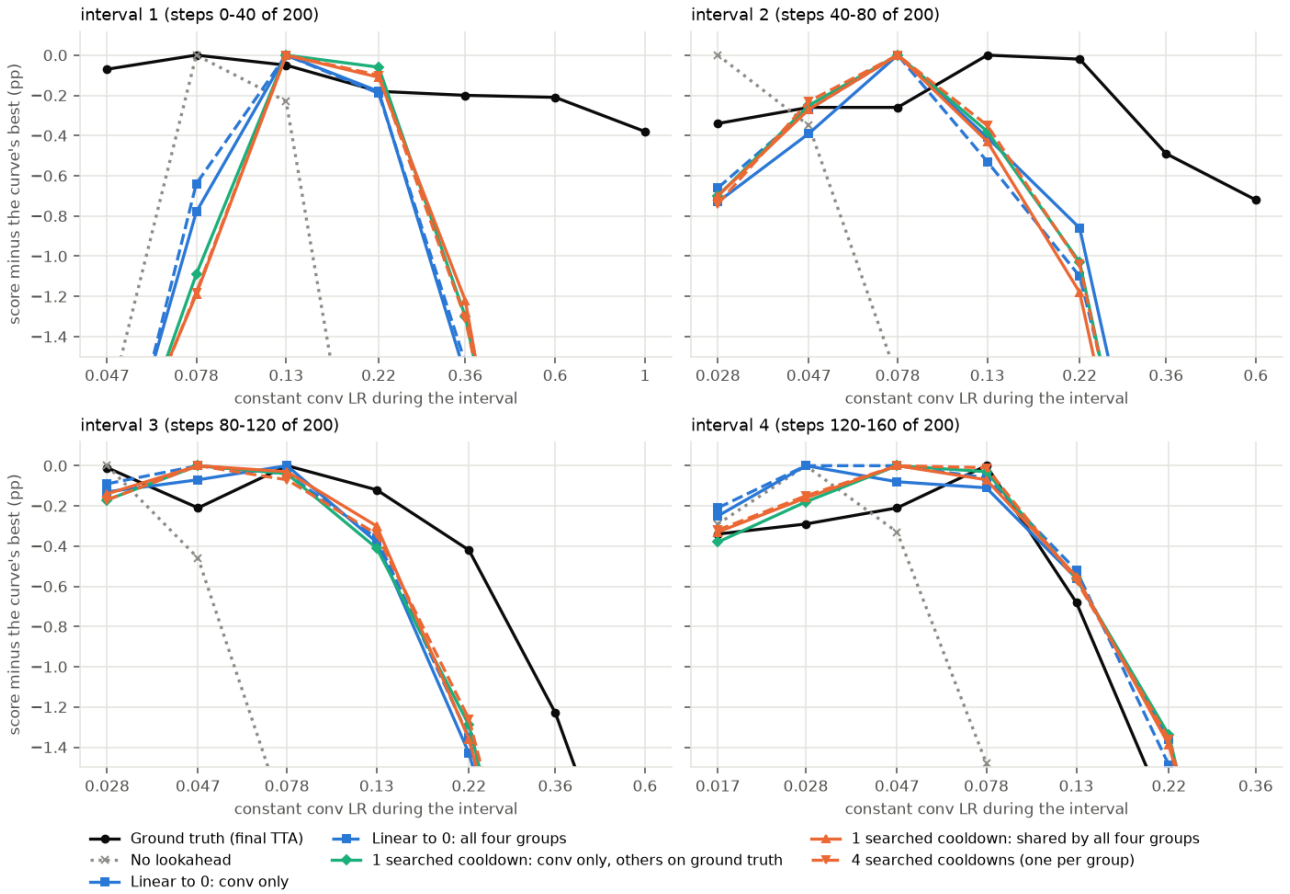



Figure 1. Each method's score per candidate, relative to its own best (the point at 0 is its pick).

### *Interval 1: score per candidate conv LR (bold: each row's best)*

| Method | 0.047 | 0.078 | 0.13 | 0.22 | 0.36 | 0.6 | 1 |
|---|---|---|---|---|---|---|---|
| Ground truth (final TTA) | 93.99 | **94.06** | 94.01 | 93.88 | 93.86 | 93.85 | 93.68 |
| No lookahead | 81.57 | **83.35** | 83.12 | 80.45 | 68.80 | 56.07 | 35.18 |
| Linear to 0: conv only | 89.04 | 90.59 | **91.37** | 91.19 | 89.78 | 87.38 | 84.39 |
| Linear to 0: all four groups | 88.85 | 90.63 | **91.27** | 91.08 | 89.74 | 87.41 | 84.10 |
| 1 searched cooldown: conv only, others on ground truth | 88.58 | 89.83 | **90.92** | 90.86 | 89.62 | 87.45 | 85.95 |
| 1 searched cooldown: shared by all four groups | 88.65 | 89.70 | **90.89** | 90.78 | 89.67 | 87.16 | 85.13 |
| 4 searched cooldowns (one per group) | 88.79 | 89.89 | **91.07** | 90.97 | 89.76 | 87.65 | 86.13 |

| Candidate | Conv only: chosen m (seeds) | Shared: chosen m (seeds) | 4 searched: chosen (conv, head, norm_bias, whiten_bias) (seeds) |
|---|---|---|---|
| 0.047 | 0.6 x1 | 0.6 x1 | (1, 1, 0.6, 0.36) x1 |
| 0.078 | 0.36 x1 | 0.36 x1 | (0.36, 1, 1, 0.22) x1 |
| 0.13 | 0.22 x1 | 0.22 x1 | (0.22, 1, 0.22, 0.22) x1 |
| 0.22 | 0.22 x1 | 0.22 x1 | (0.22, 0.36, 0.22, 1) x1 |
| 0.36 | 0.13 x1 | 0.13 x1 | (0.13, 0.047, 0.13, 0.13) x1 |
| 0.6 | 0.078 x1 | 0.078 x1 | (0.078, 1, 0.36, 0.078) x1 |
| 1 | 0.078 x1 | 0.078 x1 | (0.078, 0.6, 0.078, 0.6) x1 |

### *Interval 2: score per candidate conv LR (bold: each row's best)*

| Method | 0.028 | 0.047 | 0.078 | 0.13 | 0.22 | 0.36 | 0.6 |
|---|---|---|---|---|---|---|---|
| Ground truth (final TTA) | 93.76 | 93.84 | 93.84 | **94.10** | 94.08 | 93.61 | 93.38 |
| No lookahead | **91.12** | 90.77 | 89.51 | 87.43 | 82.45 | 71.27 | 57.08 |
| Linear to 0: conv only | 92.25 | 92.59 | **92.98** | 92.57 | 92.12 | 90.18 | 87.94 |
| Linear to 0: all four groups | 92.27 | 92.66 | **92.93** | 92.40 | 91.83 | 90.11 | 87.78 |
| 1 searched cooldown: conv only, others on ground truth | 92.16 | 92.61 | **92.86** | 92.48 | 91.83 | 89.82 | 87.93 |
| 1 searched cooldown: shared by all four groups | 92.18 | 92.61 | **92.88** | 92.45 | 91.70 | 89.73 | 87.84 |
| 4 searched cooldowns (one per group) | 92.20 | 92.71 | **92.94** | 92.59 | 91.90 | 89.87 | 87.98 |

| Candidate | Conv only: chosen m (seeds) | Shared: chosen m (seeds) | 4 searched: chosen (conv, head, norm_bias, whiten_bias) (seeds) |
|---|---|---|---|
| 0.028 | 0.6 x1 | 0.6 x1 | (0.6, 0.6, 1, 0.6) x1 |
| 0.047 | 0.36 x1 | 0.36 x1 | (0.36, 1, 0.047, 0.36) x1 |
| 0.078 | 0.22 x1 | 0.22 x1 | (0.22, 0.13, 0.22, 0.22) x1 |
| 0.13 | 0.22 x1 | 0.22 x1 | (0.22, 1, 0.36, 0) x1 |
| 0.22 | 0.13 x1 | 0.13 x1 | (0.13, 1, 1, 0.078) x1 |
| 0.36 | 0.13 x1 | 0.13 x1 | (0.13, 0.6, 0, 0.047) x1 |
| 0.6 | 0.078 x1 | 0.078 x1 | (0.078, 0.047, 0.13, 0.078) x1 |

### *Interval 3: score per candidate conv LR (bold: each row's best)*

| Method | 0.028 | 0.047 | 0.078 | 0.13 | 0.22 | 0.36 | 0.6 |
|---|---|---|---|---|---|---|---|
| Ground truth (final TTA) | 93.99 | 93.79 | **94.00** | 93.88 | 93.58 | 92.77 | 91.54 |
| No lookahead | **92.59** | 92.13 | 90.81 | 87.68 | 82.32 | 71.78 | 56.20 |
| Linear to 0: conv only | 93.26 | 93.32 | **93.39** | 93.01 | 91.96 | 90.45 | 87.63 |
| Linear to 0: all four groups | 93.28 | **93.37** | 93.33 | 93.01 | 92.02 | 90.23 | 87.59 |
| 1 searched cooldown: conv only, others on ground truth | 93.16 | **93.33** | 93.29 | 92.92 | 92.04 | 90.31 | 88.61 |
| 1 searched cooldown: shared by all four groups | 93.13 | **93.27** | 93.24 | 92.97 | 91.91 | 90.29 | 88.64 |
| 4 searched cooldowns (one per group) | 93.18 | **93.35** | 93.28 | 93.01 | 92.09 | 90.44 | 88.76 |

| Candidate | Conv only: chosen m (seeds) | Shared: chosen m (seeds) | 4 searched: chosen (conv, head, norm_bias, whiten_bias) (seeds) |
|---|---|---|---|
| 0.028 | 0.36 x1 | 0.22 x1 | (0.22, 0.13, 1, 0.22) x1 |
| 0.047 | 0.22 x1 | 0.36 x1 | (0.22, 1, 1, 0.36) x1 |
| 0.078 | 0.22 x1 | 0.22 x1 | (0.22, 1, 0.36, 0.22) x1 |
| 0.13 | 0.22 x1 | 0.22 x1 | (0.22, 1, 0.22, 0.22) x1 |
| 0.22 | 0.22 x1 | 0.13 x1 | (0.22, 0.36, 1, 0.13) x1 |
| 0.36 | 0.13 x1 | 0.13 x1 | (0.13, 0.078, 0.6, 0.13) x1 |
| 0.6 | 0.078 x1 | 0.13 x1 | (0.078, 0.36, 0.13, 0.13) x1 |

***Interval 4: score per candidate conv LR (bold: each row's best)***

| Method | 0.017 | 0.028 | 0.047 | 0.078 | 0.13 | 0.22 | 0.36 |
|---|---|---|---|---|---|---|---|
| Ground truth (final TTA) | 93.65 | 93.70 | 93.78 | **93.99** | 93.31 | 92.23 | 89.58 |
| No lookahead | 92.89 | **93.18** | 92.85 | 91.70 | 88.84 | 83.30 | 72.23 |
| Linear to 0: conv only | 93.55 | **93.80** | 93.72 | 93.69 | 93.24 | 92.44 | 90.63 |
| Linear to 0: all four groups | 93.55 | **93.76** | **93.76** | 93.70 | 93.24 | 92.27 | 90.57 |
| 1 searched cooldown: conv only, others on ground truth | 93.51 | 93.71 | **93.89** | 93.86 | 93.33 | 92.55 | 90.70 |
| 1 searched cooldown: shared by all four groups | 93.56 | 93.73 | **93.89** | 93.82 | 93.34 | 92.50 | 90.89 |
| 4 searched cooldowns (one per group) | 93.62 | 93.79 | **93.94** | 93.93 | 93.37 | 92.58 | 90.93 |

| Candidate | Conv only: chosen m (seeds) | Shared: chosen m (seeds) | 4 searched: chosen (conv, head, norm_bias, whiten_bias) (seeds) |
|---|---|---|---|
| 0.017 | 0.6 x1 | 0.6 x1 | (0.6, 1, 1, 0.6) x1 |
| 0.028 | 0.6 x1 | 0.6 x1 | (0.6, 1, 0.36, 0.6) x1 |
| 0.047 | 0.36 x1 | 0.36 x1 | (0.36, 0.22, 1, 0.36) x1 |
| 0.078 | 0.22 x1 | 0.22 x1 | (0.22, 0.047, 0.22, 0.22) x1 |
| 0.13 | 0.13 x1 | 0.13 x1 | (0.13, 0.22, 0.13, 0.13) x1 |
| 0.22 | 0.13 x1 | 0.13 x1 | (0.13, 1, 0.078, 0.13) x1 |
| 0.36 | 0.13 x1 | 0.13 x1 | (0.13, 0.078, 1, 0.13) x1 |

### *Observations*

One seed: single-run noise is about 0.13 pp, so ground-truth differences of that size are not resolved. Interval 1's ground truth is flat (93.85-94.06 from 0.047 to 0.6) and interval 3's is non-monotone (0.028: 93.99, 0.047: 93.79, 0.078: 94.00), so regrets of about 0.2 pp or less at those intervals are within noise.

All five 40-step lookahead variants pick the same LR at intervals 1 and 2, and the three searched-constant variants (conv only, shared, four per group) pick the same LR at every interval. How the other three groups cool down (continuing their ground-truth schedules, one shared multiplier, or their own multipliers) never changed the pick. Four searched multipliers score slightly higher in absolute terms than one but rank the candidates the same way.

The 40-step lookahead picks a smaller LR than the ground truth where the ground truth is clearly peaked: interval 2 (all variants pick 0.078 where 0.13 is best, 0.26 pp regret) and interval 4 (0.028-0.047 where 0.078 is best, 0.21-0.29 pp). No lookahead is worse at intervals 2 and 4 (0.34 / 0.29 pp) and always picks the smallest or second-smallest candidate after interval 1.

Linear decay and searched constants differ only at intervals 3 and 4, in opposite directions (linear conv-only is better at interval 3, searched constants at interval 4), both by about 0.2-0.3 pp: not resolved with one seed.

## Experiment 2. Hparam importance from scratch

### *Question*

How important is each of the four LRs when they are searched jointly from scratch, and does the answer depend on the main interval length?

### *Setup*

| Item | Value |
|---|---|
| Model / data | baseline_bs2000 except the four trained LRs (conv, head, norm_bias, whiten_bias), which are constant for the first L training steps from initialization |
| Main interval L | 1, 2, 5, 10, 40 steps; the score is taken right after the L steps (no lookahead) |
| Joint search | From scratch: every LR starts at 1.0. Scan each LR over x10^k, k = -3..3 (others fixed), keep the best, snap to the 0.6^k grid, then coordinate-climb on the grid (one step up / down, move while the score improves) until no LR moves: the center |
| Sweep | From the center, each LR alone over 5 grid points on each side (others at the center). Offset +1 = one grid step larger (x1/0.6), -1 = one step smaller (x0.6) |
| Seed / score | 1 seed (0); test-set TTA accuracy (%); the test cross-entropy loss is logged too |

### *Joint-search centers*

| L (steps) | Center: conv, head, norm_bias, whiten_bias | Center TTA | Evaluations |
|---|---|---|---|
| 1 | 0.13, 60, 0.00078, 60 | 25.43 | 73 |
| 2 | 0.22, 1, 1.7, 1300 | 36.89 | 71 |
| 5 | 0.078, 0.006, 2100, 1300 | 52.00 | 76 |
| 10 | 0.078, 0.017, 2100, 13 | 59.47 | 70 |
| 40 | 0.078, 2100, 280, 2100 | 84.60 | 73 |

### *Importance: TTA drop from the center (pp) at 1 step / 2 steps (mean of both sides) / worst within 5 steps*

| LR | L = 1 | L = 2 | L = 5 | L = 10 | L = 40 |
|---|---|---|---|---|---|
| conv | 0.58 / 2.01 / 16.42 | 1.00 / 2.90 / 19.69 | 3.20 / 7.96 / 32.96 | 2.02 / 4.11 / 29.37 | 1.95 / 5.26 / 49.82 |
| head | 0.03 / 0.14 / 0.29 | 0.02 / 0.02 / 0.08 | 0.10 / 0.12 / 0.18 | 0.32 / 0.49 / 0.67 | 0.54 / 1.65 / 62.56 |
| norm_bias | 0.00 / 0.00 / 0.00 | 0.05 / 0.02 / 0.08 | 0.51 / 1.46 / 28.09 | 1.12 / 2.77 / 28.84 | 0.43 / 0.84 / 6.98 |
| whiten_bias | 0.01 / 0.03 / 0.09 | 0.15 / 0.85 / 4.48 | 0.56 / 1.02 / 3.45 | 0.51 / 0.58 / 2.00 | 0.46 / 0.96 / 2.44 |

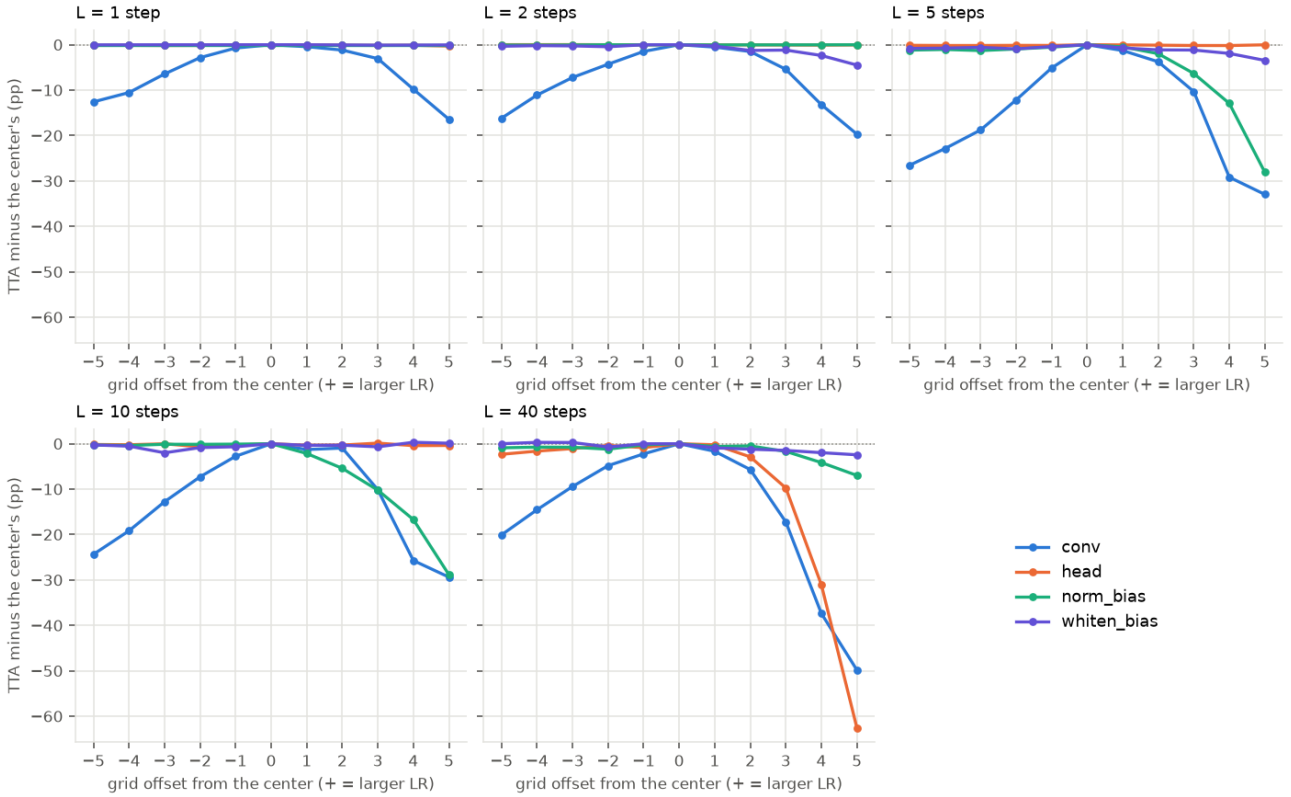



Figure 2. One-at-a-time sweeps around each L's joint-search center.

***L = 1: TTA minus the center's (pp) at each grid offset (center LR in brackets; center TTA 25.43)***

| LR | -5 | -4 | -3 | -2 | -1 | +0 | +1 | +2 | +3 | +4 | +5 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| conv (0.13) | -12.58 | -10.53 | -6.44 | -2.84 | -0.73 | +0.00 | -0.43 | -1.18 | -3.06 | -9.80 | -16.42 |
| head (60) | -0.15 | -0.14 | -0.17 | -0.16 | -0.03 | +0.00 | -0.03 | -0.11 | -0.16 | -0.12 | -0.29 |
| norm_bias (0.00078) | +0.00 | +0.00 | +0.00 | +0.00 | +0.00 | +0.00 | +0.00 | +0.00 | +0.00 | +0.00 | +0.00 |
| whiten_bias (60) | -0.03 | -0.02 | -0.01 | -0.01 | -0.01 | +0.00 | -0.01 | -0.05 | -0.09 | -0.08 | -0.04 |

***L = 2: TTA minus the center's (pp) at each grid offset (center LR in brackets; center TTA 36.89)***

| LR | -5 | -4 | -3 | -2 | -1 | +0 | +1 | +2 | +3 | +4 | +5 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| conv (0.22) | -16.19 | -11.09 | -7.21 | -4.29 | -1.48 | +0.00 | -0.52 | -1.50 | -5.41 | -13.19 | -19.69 |
| head (1) | +0.01 | +0.01 | +0.02 | +0.00 | -0.01 | +0.00 | -0.03 | -0.03 | -0.05 | -0.08 | -0.04 |
| norm_bias (1.7) | -0.07 | -0.08 | -0.05 | -0.01 | -0.04 | +0.00 | -0.05 | -0.03 | -0.03 | -0.02 | -0.01 |
| whiten_bias (1300) | -0.34 | -0.20 | -0.26 | -0.44 | -0.01 | +0.00 | -0.29 | -1.26 | -1.18 | -2.37 | -4.48 |

***L = 5: TTA minus the center's (pp) at each grid offset (center LR in brackets; center TTA 52.00)***

| LR | -5 | -4 | -3 | -2 | -1 | +0 | +1 | +2 | +3 | +4 | +5 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| conv (0.078) | -26.55 | -22.87 | -18.78 | -12.17 | -5.13 | +0.00 | -1.28 | -3.75 | -10.31 | -29.17 | -32.96 |
| head (0.006) | -0.13 | -0.17 | -0.17 | -0.14 | -0.16 | +0.00 | -0.04 | -0.10 | -0.17 | -0.18 | -0.02 |
| norm_bias (2100) | -1.19 | -1.04 | -1.29 | -0.94 | -0.54 | +0.00 | -0.47 | -1.97 | -6.33 | -12.92 | -28.09 |
| whiten_bias (1300) | -0.81 | -0.75 | -0.58 | -0.91 | -0.41 | +0.00 | -0.71 | -1.13 | -1.17 | -1.92 | -3.45 |

***L = 10: TTA minus the center's (pp) at each grid offset (center LR in brackets; center TTA 59.47)***

| LR | -5 | -4 | -3 | -2 | -1 | +0 | +1 | +2 | +3 | +4 | +5 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| conv (0.078) | -24.35 | -19.17 | -12.77 | -7.28 | -2.76 | +0.00 | -1.28 | -0.95 | -10.13 | -25.72 | -29.37 |
| head (0.017) | -0.17 | -0.23 | -0.02 | -0.67 | -0.31 | +0.00 | -0.34 | -0.30 | +0.13 | -0.42 | -0.38 |
| norm_bias (2100) | -0.29 | -0.39 | -0.12 | -0.13 | -0.09 | +0.00 | -2.15 | -5.42 | -10.23 | -16.70 | -28.84 |
| whiten_bias (13) | -0.26 | -0.53 | -2.00 | -0.85 | -0.67 | +0.00 | -0.36 | -0.32 | -0.68 | +0.32 | +0.11 |

***L = 40: TTA minus the center's (pp) at each grid offset (center LR in brackets; center TTA 84.60)***

| LR | -5 | -4 | -3 | -2 | -1 | +0 | +1 | +2 | +3 | +4 | +5 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| conv (0.078) | -20.08 | -14.54 | -9.42 | -4.83 | -2.24 | +0.00 | -1.67 | -5.69 | -17.22 | -37.36 | -49.82 |
| head (2100) | -2.33 | -1.63 | -1.07 | -0.40 | -0.85 | +0.00 | -0.23 | -2.90 | -9.74 | -31.04 | -62.56 |
| norm_bias (280) | -0.90 | -0.75 | -0.77 | -1.20 | -0.24 | +0.00 | -0.62 | -0.48 | -1.69 | -4.15 | -6.98 |
| whiten_bias (2100) | +0.00 | +0.30 | +0.27 | -0.72 | -0.01 | +0.00 | -0.91 | -1.21 | -1.47 | -1.97 | -2.44 |

## Experiment 3. Representation quality of schedule families

### *Question*

How close to the baseline can each family of conv LR schedules get, under no transform and under the ramp, reverse ramp and wave transforms, when searched by the same global search?

### *Setup*

| Item | Value |
|---|---|
| Model / data | baseline_bs2000; only the conv LR schedule is searched, the other groups stay on the baseline. The transform multiplies the conv LR the search specifies |
| Conditions | No transform; ramp x 4^(2x-1) (x0.25 -> x4); reverse ramp x 4^(1-2x) (x4 -> x0.25); wave x 2^sin(2 pi x) (x1 -> x2 -> x0.5 -> x1); x = fraction of training |
| Families | 5 constant intervals (5 values); linear decay to 0 (initial LR); power decay lr x (1-t)^p (lr, p; p = 1 is linear); 2 knots at t = 0, 0.5; 3 knots at t = 0, 1/3, 2/3; 5 knots at t = 0, 0.2, 0.4, 0.6, 0.8 (knot values; linear between knots, 0 at the end) |
| Search | Generalized global_neighbour (global_family): coordinate search over the family's parameters on multiplicative grids (LR values x0.6, p x0.8), one grid step at a time, moving while the score improves; every candidate is a full training run on seed 0 from the same initial state |
| Start | The oracle spec (baseline conv schedule divided by the transform) projected onto the family: interval means or knot values snapped to the grid; linear / power start at 0.22 with p = 1 |
| Evaluation | Start and best schedules retrained on held-out seeds 100-107 (mean test TTA; not part of the search) |

### *References (held-out TTA, seeds 100-107)*

| | No transform | Ramp | Reverse ramp | Wave |
|---|---|---|---|---|
| Baseline linear decay, no transform (the target) | 93.92 | 93.92 | 93.92 | 93.92 |
| Baseline spec kept under the transform (do-nothing) | 93.92 | 93.42 | 93.57 | 93.72 |

### *Held-out TTA of the best schedule (start = the projected oracle before search; search cost)*

| Family | No transform | Ramp | Reverse ramp | Wave |
|---|---|---|---|---|
| 5 constant intervals | **93.62** (start 93.52; 18 runs, 182 s) | **93.65** (start 93.32; 28 runs, 285 s) | **93.86** (start 93.81; 28 runs, 279 s) | **93.82** (start 93.11; 22 runs, 222 s) |
| Linear decay (1 param) | **93.86** (start 93.92; 3 runs, 31 s) | **93.42** (start 93.42; 3 runs, 31 s) | **93.59** (start 93.57; 4 runs, 41 s) | **93.91** (start 93.72; 3 runs, 31 s) |
| Power decay (lr, p) | **93.86** (start 93.92; 8 runs, 81 s) | **93.95** (start 93.42; 14 runs, 141 s) | **93.59** (start 93.57; 6 runs, 61 s) | **93.91** (start 93.72; 5 runs, 51 s) |
| 2 knots (t = 0, 0.5) | **93.86** (start 93.86; 5 runs, 51 s) | **93.77** (start 93.14; 9 runs, 91 s) | **93.88** (start 93.85; 8 runs, 83 s) | **93.83** (start 93.72; 12 runs, 121 s) |
| 3 knots (t = 0, 1/3, 2/3) | **93.86** (start 93.94; 12 runs, 121 s) | **93.68** (start 93.74; 10 runs, 101 s) | **93.78** (start 93.90; 13 runs, 131 s) | **94.06** (start 93.90; 7 runs, 73 s) |
| 5 knots (t = 0, 0.2, ..., 0.8) | **93.94** (start 93.97; 13 runs, 131 s) | **93.96** (start 93.91; 17 runs, 170 s) | **93.91** (start 93.91; 11 runs, 111 s) | **93.94** (start 93.97; 11 runs, 111 s) |

### *Best parameter values (LR values in order; power: lr, p)*

| Family | No transform | Ramp | Reverse ramp | Wave |
|---|---|---|---|---|
| 5 constant intervals | 0.13, 0.22, 0.078, 0.047, 0.028 | 0.36, 0.13, 0.078, 0.028, 0.0036 | 0.028, 0.078, 0.13, 0.078, 0.047 | 0.078, 0.078, 0.13, 0.078, 0.017 |
| Linear decay (1 param) | 0.13 | 0.22 | 0.078 | 0.13 |
| Power decay (lr, p) | 0.13, 0.64 | 0.36, 2 | 0.078, 1 | 0.13, 1 |
| 2 knots (t = 0, 0.5) | 0.22, 0.13 | 0.6, 0.078 | 0.078, 0.078 | 0.13, 0.078 |
| 3 knots (t = 0, 1/3, 2/3) | 0.13, 0.22, 0.047 | 0.36, 0.13, 0.047 | 0.078, 0.078, 0.047 | 0.13, 0.078, 0.13 |
| 5 knots (t = 0, 0.2, ..., 0.8) | 0.13, 0.13, 0.13, 0.078, 0.047 | 1, 0.36, 0.22, 0.047, 0.017 | 0.047, 0.078, 0.078, 0.13, 0.13 | 0.13, 0.078, 0.078, 0.13, 0.078 |

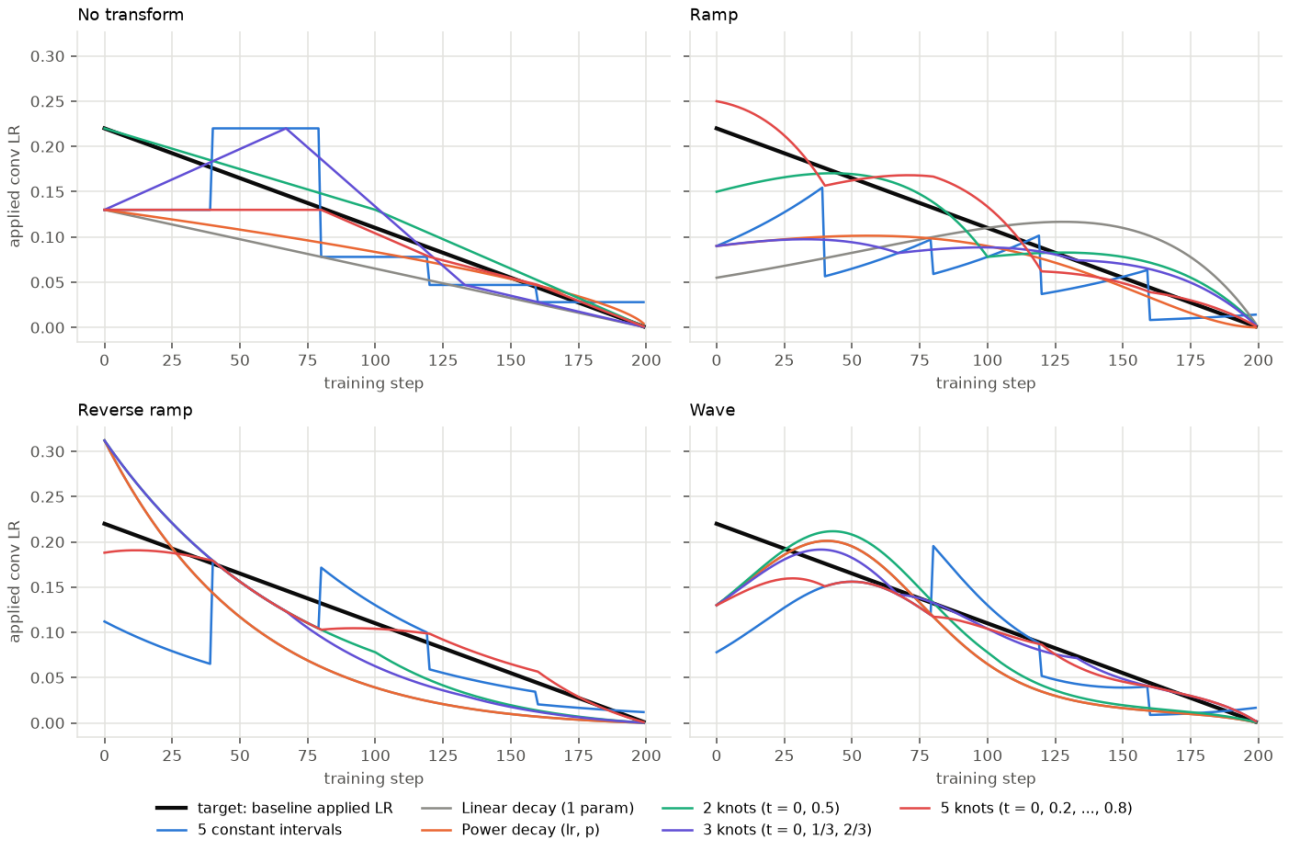



Figure 3. Applied conv LR (spec x transform) of each family's best schedule; the black line is the baseline's applied LR, which a perfect search would reproduce.

### *Observations*

Held-out means over 8 seeds have a standard error of about 0.045 pp, so differences below about 0.1 pp are not resolved. Target: the baseline's 93.92 on the same seeds.

5 knots is the only family that reached the target in all four conditions in these searches (93.91-93.96), and its start, the oracle projected onto the family, is already there (93.91-93.97): 5 knots on the grid represent all four target schedules.

5 constant intervals of conv reach only 93.62 here without a transform (start 93.52), but that is this search's local optimum, not the family's limit. Earlier 5-interval searches at batch size 2000 (same held-out seeds) did better: conv only 93.77 (interval search; conv 0.13, 0.13, 0.078, 0.047, 0.01) and 93.72 (global_interval), and with all four LRs as 5 intervals 93.92 (interval search), matching the baseline, and 93.85 (global_interval). This search kept the last interval at 0.028 where those went to 0.01-0.017: a single-seed coordinate search can stop one grid step short at the end. Under the transforms it reaches 93.65 / 93.86 / 93.82 (ramp / reverse ramp / wave; projected starts 93.32 / 93.81 / 93.11).

Linear decay matches the target without a transform (93.86-93.92) and under wave (93.91), but no initial LR compensates a ramp (93.42) or a reverse ramp (93.59): the shape is wrong, not the scale. Power decay fixes the ramp (93.95 with p = 2, lr 0.36: decaying faster) but not the reverse ramp (93.59): (1-t)^p cannot rise early, which the reverse ramp needs.

2 and 3 knots land in between (93.68-94.06), with no consistent ordering across conditions.

Single-seed search sometimes leaves a good start for a worse held-out schedule (no transform: linear 93.92 -> 93.86, 3 knots 93.94 -> 93.86, 5 constant intervals moved its first value from 0.22 to 0.13): the search optimizes seed 0's test score, the same winner's curse as before.

## Experiment 4. Hparam range from scratch (Experiment 2 redone)

### *Question*

Experiment 2's +-5 sweeps depend on where the center landed. How far can each LR move from the center, on each side, before the score drops by 10%? A range that ends on both sides marks an LR with an optimum; a side that never drops marks an LR the score is insensitive to in that direction.

### *Setup*

| Item | Value |
|---|---|
| Model / score | As in Experiment 2: baseline_bs2000 except the four LRs, constant for the first L steps from initialization; test TTA right after the L steps; 1 seed (0) |
| Center | Experiment 2's joint-search center for each L (re-evaluated; it reproduces Experiment 2's TTA) |
| Probe | One LR at a time (others at the center), one 0.6^k grid step at a time on each side (+1 = x1/0.6, larger), until TTA <= 0.9 x the center's TTA (a 10% relative drop) or 20 steps. A diverged run counts as a drop |

### *Stop offsets (smaller side / larger side) and the LR range within 10% of the center*

| LR | L = 1 | L = 2 | L = 5 | L = 10 | L = 40 |
|---|---|---|---|---|---|
| conv | -2 / +3<br>0.078 - 0.36 | -2 / +3<br>0.13 - 0.6 | -2 / +3<br>0.047 - 0.22 | -2 / +3<br>0.047 - 0.22 | -3 / +3<br>0.028 - 0.22 |
| head | <-20 / +7<br>0.0022 - 1300 | <-20 / +14<br>3.7e-05 - 770 | <-20 / >20<br>2.2e-07 - 170 | <-20 / >20<br>6.1e-07 - 460 | <-20 / +3<br>0.078 - 5900 |
| norm_bias | <-20 / >20<br>2.9e-08 - 21 | <-20 / +17<br>6.1e-05 - 5900 | <-20 / +3<br>0.078 - 5900 | <-20 / +3<br>0.078 - 5900 | <-20 / +6<br>0.01 - 3500 |
| whiten_bias | <-20 / +10<br>0.0022 - 5900 | <-20 / +5<br>0.047 - 9800 | <-20 / +6<br>0.047 - 16000 | <-20 / +17<br>0.00047 - 46000 | <-20 / +11<br>0.078 - 350000 |

<-20 / >20: no 10% drop within 20 steps on that side (the range shown then ends at the last probed point). Center TTA: L = 1: 25.43, L = 2: 36.89, L = 5: 52.00, L = 10: 59.47, L = 40: 84.60.

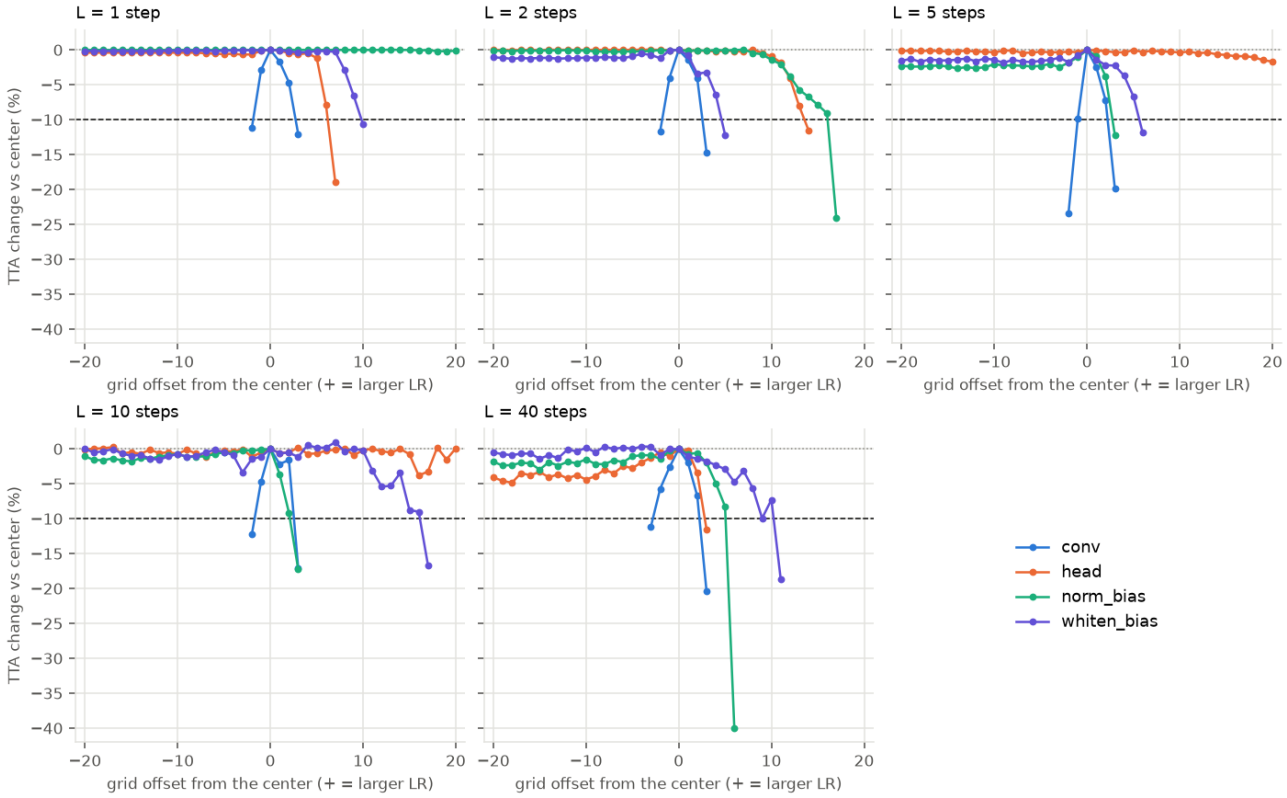



Figure 4. Relative TTA change along each probe; the dashed line is the 10% stop. Points below -40% are drawn at -40%.

### *Observations*

One seed: single-run noise is about 0.13 pp at the full 200 steps; the stop criterion (a 10% relative drop: 2.5 pp at L = 1, 8.5 pp at L = 40) is far above it, so the stop offsets are not noise.

Conv is the only LR with an optimum: its range ends on both sides at every L, at -2 or -3 grid steps (smaller) and +3 (larger). The LR range within 10% of the center spans 5-6 grid points, a factor of 4.6-7.9 in LR (0.078-0.36 at L = 1, 0.028-0.22 at L = 40).

Head, norm_bias and whiten_bias never drop by 10% on the smaller side within 20 steps (down to 0.6^20 = 3.7e-5 x the center) at any L. The worst smaller-side loss is 4.10 pp (head, L = 40); most are below 1.5 pp, and at L = 1-2 below 0.5 pp. These three LRs only have an upper limit.

Their larger sides do end, but late and without a pattern across L: head at +7 / +14 / none / none / +3 (L = 1 / 2 / 5 / 10 / 40), norm_bias at none / +17 / +3 / +3 / +6, whiten_bias at +10 / +5 / +6 / +17 / +11. So their Experiment 2 centers mostly mark where "too large" begins, not an optimum: any value below that edge scores about the same.

At L = 10 a few probe points beat the center (whiten_bias +0.57 pp, head +0.15 pp): the joint-search center is not an exact optimum on one seed, which is the center dependence this experiment was meant to remove.

## Experiment 5. Which non-conv interval schedules help

### *Question*

The two original 5-interval searches scored 93.77 (conv only) and 93.92 (all four LRs) on held-out seeds, but their conv schedules differ in the last two intervals (0.078, 0.017 vs 0.047, 0.01). With conv fixed, do the other groups' searched intervals help, and which group?

### *Setup*

| Item | Value |
|---|---|
| Schedules | C4 = conv constant 0.13, 0.13, 0.078, 0.047, 0.01 (5 x 40 steps; the four-LR interval search's conv). Searched intervals: head 2100, 2100, 2100, 2100, 1300; norm_bias 770, 770, 770, 460, 280; whiten_bias 460, 460, 280, 280, 280. Baseline: conv 0.22 -> 0, head 1300 -> 0, norm_bias 99 -> 0 (linear over 200 steps), whiten_bias 36 -> 0 over 75 steps, then 0 |
| Runs | Plain training runs (no search) on held-out seeds 100-107, current code; the paired difference is per seed against the first row |

### *Held-out TTA*

| Conv | head | norm_bias | whiten_bias | Held-out TTA (mean +- SE) | vs first row (paired) |
|---|---|---|---|---|---|
| C4 | baseline | baseline | baseline | 93.95 +- 0.06 | - |
| C4 | searched | baseline | baseline | 93.90 +- 0.05 | -0.05 +- 0.08 |
| C4 | baseline | searched | baseline | 93.77 +- 0.04 | -0.18 +- 0.08 |
| C4 | baseline | baseline | searched | 93.88 +- 0.05 | -0.07 +- 0.07 |
| C4 | searched | searched | searched | 93.92 +- 0.04 | -0.03 +- 0.07 |
| baseline (0.22 -> 0) | searched | searched | searched | 93.87 +- 0.04 | -0.08 +- 0.07 |
| baseline (0.22 -> 0) | baseline | baseline | baseline | 93.92 +- 0.05 | -0.03 +- 0.07 |

For reference, the original conv-only interval search's schedule (conv 0.13, 0.13, 0.078, 0.078, 0.017, the others on the baseline) scored 93.77 on the same seeds (older code).

### *Observations*

The other groups' searched intervals do not help. With conv fixed at C4 and the other three groups on the baseline, the held-out TTA is 93.95 +- 0.06, as high as any row. Switching all three to their searched intervals gives 93.92 (-0.03 +- 0.07 paired); switching one at a time gives head -0.05 +- 0.08, whiten_bias -0.07 +- 0.07 and norm_bias -0.18 +- 0.08 (about 2.3 SE: norm_bias's searched intervals, 770 -> 280 instead of 99 -> 0, probably hurt).

The 93.77 vs 93.92 gap between the original conv-only and four-LR searches came from conv, not from the other groups. Both conv schedules with the other groups on the baseline: 0.13, 0.13, 0.078, 0.078, 0.017 scores 93.77 and C4 (0.13, 0.13, 0.078, 0.047, 0.01) 93.95, +0.18 +- 0.05 paired (7 of 8 seeds). The four-LR rerun reproduces the original run's per-seed scores exactly, so the older 93.77 run is directly comparable.

So 5 constant intervals of conv alone do match the baseline (93.95 vs the baseline's 93.92 on these seeds) when the last two intervals are low enough (0.047, 0.01); Experiment 3's conv-only search stopped at 0.047, 0.028.

The searched non-conv intervals with the baseline's linear conv score 93.87 (-0.08 +- 0.07 vs the C4 control), close to the baseline itself: the other groups' schedules matter little with either conv schedule.

## Experiment 6. Seed noise of the baseline

### *Question*

How much does the baseline's score vary across training seeds, and how many seeds does a comparison need?

### *Setup*

| Item | Value |
|---|---|
| Runs | baseline_bs2000, plain training runs on seeds 0-127 (seed 0 is the search seed; 100-107 are the held-out seeds); the seed sets the initialization, data order and augmentation |
| Score | Test TTA accuracy (%) and test accuracy without TTA |

### *Baseline across seeds*

| Metric | Seeds | Mean | SD (one run) | SE of the mean | Min | Max |
|---|---|---|---|---|---|---|
| TTA | 128 | 93.942 | 0.121 | 0.011 | 93.61 | 94.28 |
| No TTA | 128 | 93.164 | 0.140 | 0.012 | 92.69 | 93.52 |

### *Averaging over seeds*

| Seeds per mean (n) | SD of an n-seed mean: SD/sqrt(n) | Measured (disjoint blocks of n) | Smallest difference two n-seed means resolve (95%) |
|---|---|---|---|
| 1 | 0.121 | 0.121 (128 blocks) | 0.34 pp |
| 2 | 0.086 | 0.089 (64 blocks) | 0.24 pp |
| 4 | 0.061 | 0.069 (32 blocks) | 0.17 pp |
| 8 | 0.043 | 0.051 (16 blocks) | 0.12 pp |
| 16 | 0.030 | 0.041 (8 blocks) | 0.08 pp |
| 32 | 0.021 | 0.028 (4 blocks) | 0.06 pp |

Seed 0 (the search seed): 93.92. Held-out seeds 100-107: mean 93.92. All seeds: 93.94.

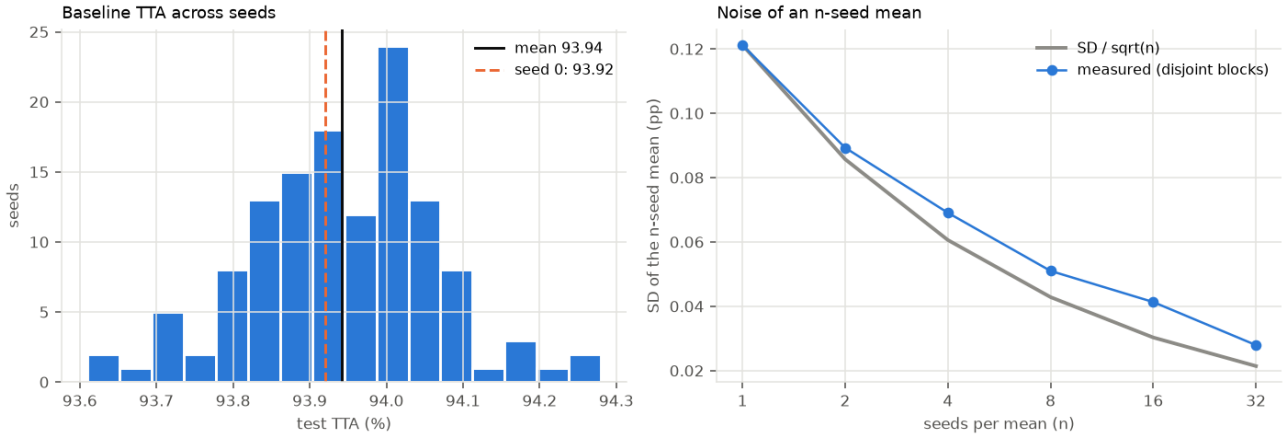



Figure 5. Left: the baseline's TTA over seeds. Right: the spread of an n-seed mean, predicted from the single-run SD and measured on disjoint blocks of seeds.

### *Observations*

One run of the baseline has a seed SD of 0.121 pp in TTA (128 seeds: mean 93.94, range 93.61-94.28, 0.67 pp from worst to best seed); without TTA the SD is 0.140 pp (mean 93.16). The mean over all 128 seeds is known to +-0.011 pp (SE).

Averaging over seeds shrinks the noise about as SD / sqrt(n): measured on disjoint blocks of seeds, an 8-seed mean has SD 0.051 (predicted 0.043; 16 blocks, so the measured value itself is uncertain by about +-20%). Telling two schedules apart at 95% takes a difference of about 0.34 pp with 1 seed each, 0.12 pp with 8 seeds and 0.06 pp with 32 seeds (unpaired; pairing by seed helps only if the seeds' effects are shared).

The seeds used so far are typical: seed 0 (the search seed) scores 93.92 and the held-out seeds 100-107 average 93.92, both within 0.02 pp of the 128-seed mean.

## Experiment 7. Hparam range around the baseline

### *Question*

Starting from the baseline instead of a from-scratch center, how far can each LR's initial value move, on each side, before the full run's score drops by 10%?

### *Setup*

| Item | Value |
|---|---|
| Runs | baseline_bs2000, full 200-step runs, seed 0. One LR at a time (the others on the baseline) |
| Probe | The LR's whole baseline schedule is scaled so its initial value steps along the 0.6^k grid (+1 = x1/0.6, larger), on each side until TTA <= 0.9 x the baseline's TTA on seed 0, or 20 steps. A diverged run counts as a drop |

### *Stop offsets (baseline TTA on seed 0: 93.92; stop at 84.53)*

| LR | Baseline initial LR | Stop (smaller / larger) | Initial LR range within 10% |
|---|---|---|---|
| conv | 0.22 | -6 / +6 | 0.017 - 2.8 (x165) |
| head | 1300 | <-20 / +8 | 0.047 - 46000 (x9.79e+05) |
| norm_bias | 99 | <-20 / +9 | 0.0036 - 5900 (x1.64e+06) |
| whiten_bias | 36 | <-20 / >20 | 0.0013 - 980000 (x7.54e+08) |

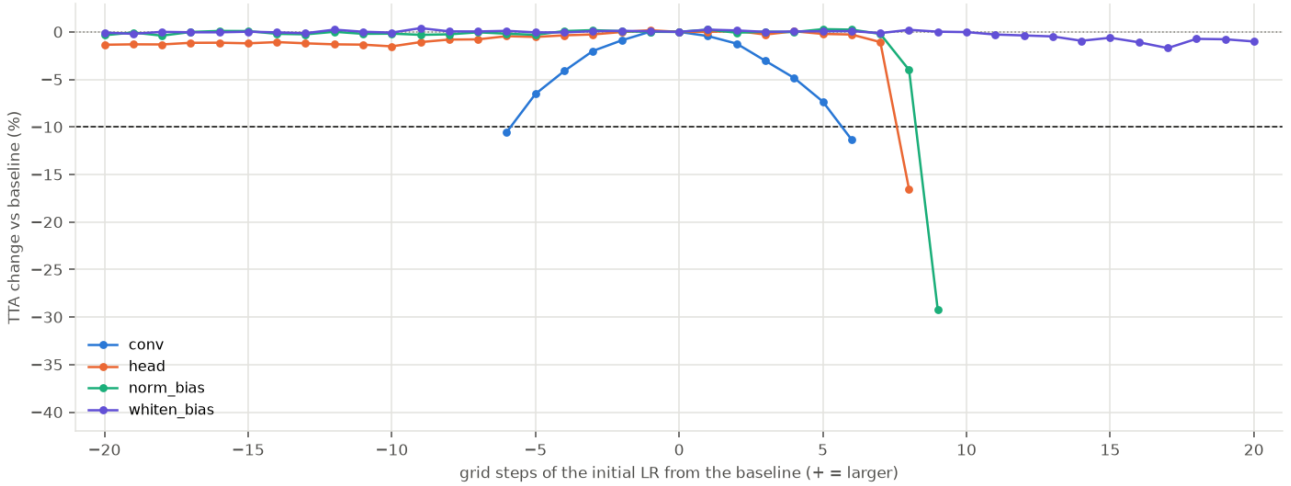



Figure 6. Relative TTA change of full runs as one LR's initial value moves from the baseline; dashed: the 10% stop. Points below -40% are drawn at -40%.

### *95% confidence set for the best initial LR*

Each probed value is one run on seed 0, so its score carries the single-run noise of Experiment 6 (SD 0.121 pp, assumed the same at every value and independent between values). A value is kept when its score is within c of the best observed score, with c = q x SD and q the 95% quantile of max_j (Z_j - Z_1) over the k probed values (Z independent standard normals; Monte Carlo). This is Gupta's subset selection: the set contains the best probed value with probability at least 95%, even in the hardest case where all values are equally good. (open) marks an end that reaches the probe limit on a side without a 10% drop, so the set may continue beyond it.

| LR | Probed points k | Cutoff c (pp) | Best observed | 95% set (initial LR) | Points in set |
|---|---|---|---|---|---|
| conv | 13 | 0.43 | 0.13 (93.95) | 0.13 - 0.22 | 2 of 13 |
| head | 29 | 0.47 | 770 (94.08) | 280 - 27000 | 10 of 29 |
| norm_bias | 30 | 0.47 | 1300 (94.19) | (open) 0.006 - 3500 | 22 of 30, with gaps |
| whiten_bias | 41 | 0.48 | 0.36 (94.29) | (open) 0.0013 - 5900 | 28 of 41, with gaps |

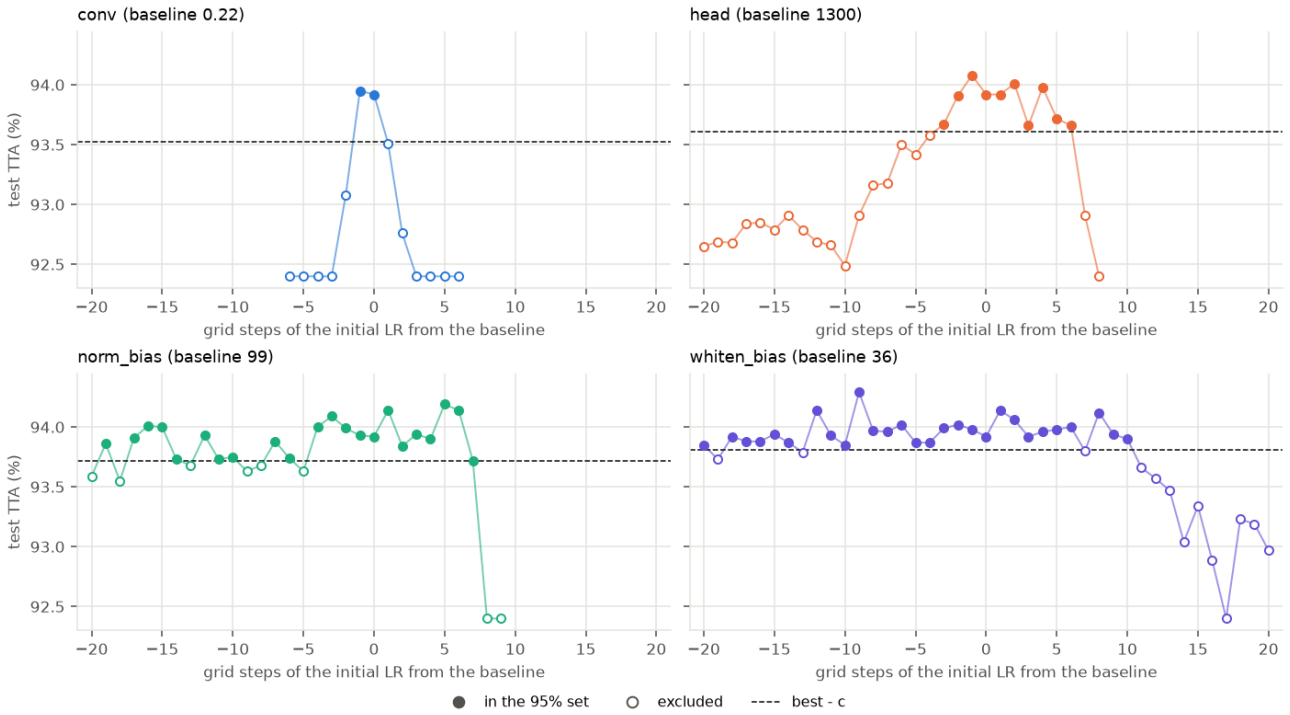



Figure 7. The probes near the top: filled points are in the 95% set (score at least best - c, dashed); points below 92.4 are drawn at 92.4.

### *Observations*

One seed (0); the 10% stop (84.53) is about 80 single-run SDs (Experiment 6) below the baseline, so the stop offsets are not noise; differences under about 0.34 pp between single points are.

Conv has a clear optimum, and the baseline sits on it: 0.13 scores 93.95 and 0.22 (baseline) 93.92; one step further costs 0.4-0.8 pp (0.36: 93.51; 0.078: 93.08). Its 10% range ends at -6 (0.01: 84.03) and +6 (4.6: 83.28), so in full runs conv tolerates a factor of about 165 (0.017 to 2.8) before losing 10%, much wider than in the from-scratch L-step setting of Experiment 4 (5-6 grid points).

Head, norm_bias and whiten_bias never drop by 10% on the smaller side within 20 steps (down to 3.7e-5 x the baseline). Norm_bias and whiten_bias are flat there (all within 93.55-94.29, within about 0.4 pp of the baseline, roughly the single-run noise range); head loses about 1.2 pp once it is below about 8 (92.5-92.9 from 7.7 down to 0.047), still far from 10%.

On the larger side, head stops at +8 (76000: 78.35; 46000 still 92.91) and norm_bias at +9 (9800: 66.44; 5900: 90.16), both after 6-7 steps of no loss: they fail by blowing up, not gradually. whiten_bias never stops: up to 980000 (x27000) it stays above 92.3.

So around the baseline only conv needs tuning; the three other LRs only need to stay below a blow-up edge about 35x above their baseline values (head, norm_bias; the last points without loss are 46000 and 3500), and whiten_bias barely matters at all.

95% confidence sets for the best initial LR (single runs with SD 0.121 pp; cutoff c = 0.43-0.48 pp below the best observed score, wider for more probed points). Conv: 0.13-0.22, the two values at the top (93.95, 93.92); the next ones are excluded (0.36: 93.51, only 0.01 pp under the cutoff 93.52; 0.078: 93.08), so the best conv initial LR is pinned to two grid points, and the baseline's 0.22 is one of them.

Head: 280-27000 (10 of 29 points, contiguous), a factor of about 100 around the best observed 770; below 280 the scores fall off (170: 93.58), and 46000 is 1.0 pp below the baseline (92.91). The baseline's 1300 is inside.

Norm_bias: anything from the smallest probed values (0.006; open, as the probe never dropped 10% on that side) up to 3500, with gaps: the scores below 3500 are within noise of each other, so the data do not locate a best value, only an upper edge. Whiten_bias: likewise from 0.0013 (open) up to 5900; from 9800 upward the scores fall (93.66 and lower) and are excluded.

## Experiment 8. Largest LR within the margin, per L (from Experiment 4)

### *Question*

The interval search picks each LR in its first interval as the largest LR scoring within a margin of the best (directional_search with initial_lr_search, initial_dropoff_margin = 0.02). Applied at main interval lengths L = 1, 2, 5, 10, 40 from scratch, does the pick at L = 1 predict the pick at L = 40?

### *Setup*

| Item | Value |
|---|---|
| Data | Experiment 4's probes (no new runs): for each L, one LR at a time on the 0.6^k grid around Experiment 2's center, the other three LRs at the center, TTA right after L steps, seed 0. The probes extend past every point where the rule stops |
| Rule | As in train_cifar.py: from the center, step up while the score stays within the margin of the best so far, then down the same way (at most 20 steps per side); pick the largest LR within the margin of the best. Margin 0.02 TTA = 2 pp (the code's value); 1 and 0.5 pp shown too |

### *Picks with the code's margin (2 pp): LR (grid steps from the center), and where it lies relative to Experiment 7's full-run 95% range for the best initial LR (last row)*

| L (steps) | Center TTA | conv | head | norm_bias | whiten_bias |
|---|---|---|---|---|---|
| 1 | 25.43 | 0.36 (+2), above | 770 (+5), in | 21 (+20)*, in | 5900 (+9), in |
| 2 | 36.89 | 0.6 (+2), above | 460 (+12), in | 770 (+12), in | 5900 (+3), in |
| 5 | 52.00 | 0.13 (+1), in | 170 (+20)*, below | 5900 (+2), above | 9800 (+4), above |
| 10 | 59.47 | 0.22 (+2), in | 36 (+15), below | 2100 (+0), in | 2100 (+10), in |
| 40 | 84.60 | 0.13 (+1), in | 3500 (+1), in | 1300 (+3), in | 9800 (+3), above |
| Exp. 7 | full run | 0.13 - 0.22 | 280 - 27000 | (open) 0.006 - 3500 | (open) 0.0013 - 5900 |

* the upward probe reached the 20-step limit without falling out of the margin, so the pick is the probe limit. in / above / below: inside, above or below the Experiment 7 range ((open): that end of the range is not bounded by the data). Experiment 7 is a different context (full 200-step runs, the other LRs on the baseline schedules), the target a short-horizon pick should predict.

### *Conv pick by margin*

| L (steps) | conv, margin 2 pp | conv, margin 1 pp | conv, margin 0.5 pp |
|---|---|---|---|
| 1 | 0.36, above | 0.22, in | 0.22, in |
| 2 | 0.6, above | 0.36, above | 0.22, in |
| 5 | 0.13, in | 0.078, below | 0.078, below |
| 10 | 0.22, in | 0.078, below | 0.078, below |
| 40 | 0.13, in | 0.078, below | 0.078, below |
| Exp. 7 | 0.13 - 0.22 | 0.13 - 0.22 | 0.13 - 0.22 |

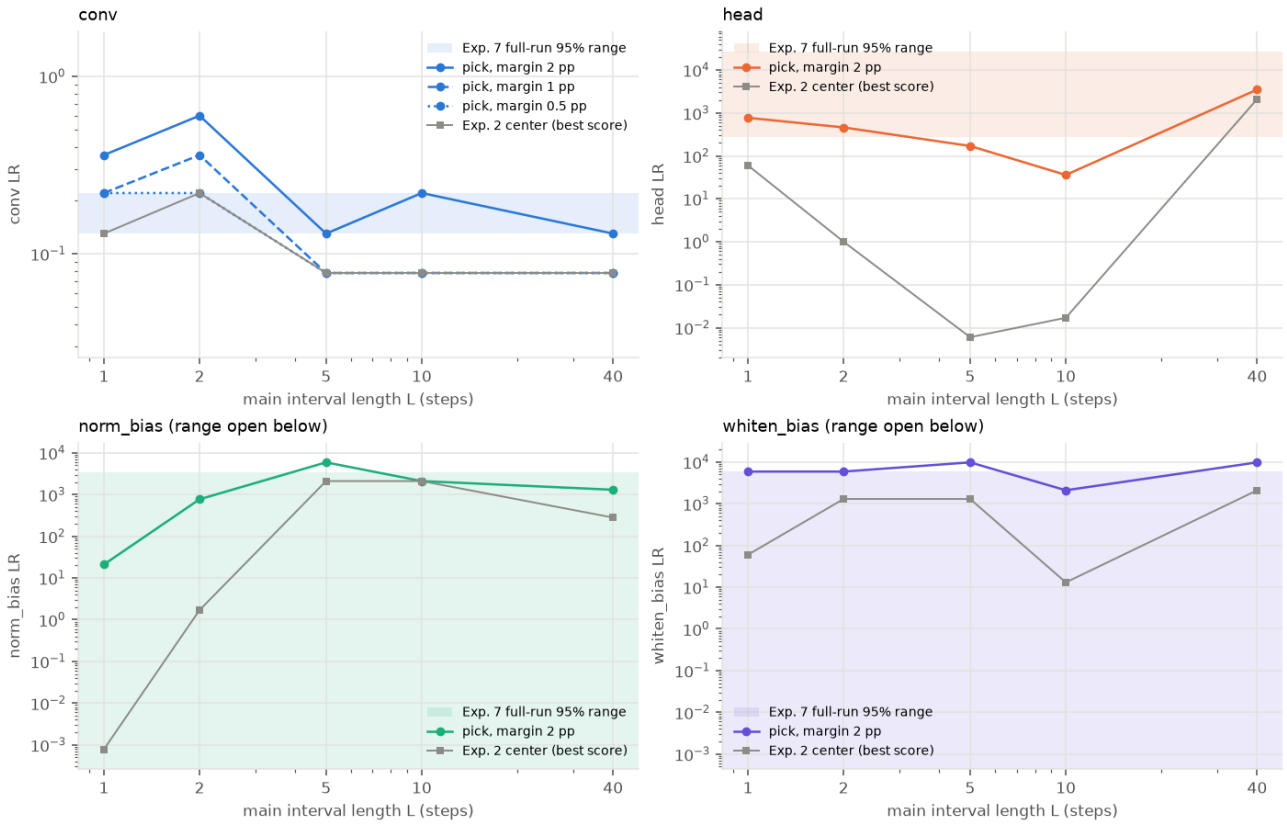



Figure 8. The rule's pick per L (conv at three margins, the others at the code's 2 pp), Experiment 2's best-score center, and the shaded reference: Experiment 7's full-run 95% range for the best initial LR.

### *Observations*

Reference: Experiment 7's full-run 95% ranges for the best initial LR (conv 0.13-0.22, head 280-27000, norm_bias up to 3500, whiten_bias up to 5900; the last two open below, so for them only "above" is a real miss). A good short-horizon pick lands inside.

Conv: with the code's 2 pp margin, L = 1 and L = 2 pick above the range (0.36, 0.6) and L = 5, 10 and 40 inside it (0.13, 0.22, 0.13). So L = 1 does not predict L = 40's pick (2 grid steps apart, 0.36 vs 0.13), and the long-horizon picks are the ones that match the full run. With tighter margins it reverses: at 1 pp and 0.5 pp, L = 1 picks 0.22 (inside) and L = 5-40 pick 0.078 (one step below the range). No single margin puts every L inside.

Why the margin interacts with L: it is absolute, but the score scale is not (the centers score 25.43 at L = 1 and 84.60 at L = 40), and at every L the 2 pp pick sits 1-2 grid steps above that L's best-score center (L = 1: 0.13 -> 0.36; L = 40: 0.078 -> 0.13). The best-score centers alone are inside the range at L = 1 and 2 (0.13, 0.22) and one step below it at L >= 5 (0.078): short horizons favour slightly larger conv LRs than long ones, and the 2 pp rule's upward step compensates at L >= 5 but overshoots at L <= 2.

Head: L = 1 (770), 2 (460) and 40 (3500) pick inside the range; L = 5 (170) and 10 (36) below it, by 1-4 grid steps although the upward probe at L = 5 hit the 20-step limit. norm_bias and whiten_bias: their ranges are open below, so the only misses are picks above the upper edge: norm_bias at L = 5 (5900 > 3500), whiten_bias at L = 5 and 40 (9800 > 5900), each one grid step over. L = 1's picks (21 and 5900) are inside, but their ranges are wide enough that this says little.

So in this data L = 1 lands inside the full-run range for all three of head, norm_bias and whiten_bias (L = 40: two of three), but for conv, the one LR with a narrow range, L = 1 overshoots with the code's margin, while L >= 5 lands inside.

Caveat: in each L the other three LRs sit at that L's own Experiment 2 center, which differs between L (e.g. head 60 at L = 1, 0.006 at L = 5, 2100 at L = 40), so the comparison across L is not in a fixed context, and Experiment 7's context (full runs, baseline schedules) differs from all of them. Running the rule from a common start at every L would remove the first confound; it needs new runs.

## Experiment 9. Representation quality of schedule families, noise-aware search

### *Question*

Experiment 3's search scored every candidate on one run (seed 0), so it climbed on seed noise and stopped at schedules that were lucky on seed 0 (e.g. 5 constant intervals: 93.99 on seed 0, 93.62 held-out). With a search that only moves to a neighbour that is statistically better given the seed noise, and adds seeds until it can tell, how close does each family get?

### *Setup*

| Item | Value |
|---|---|
| As Experiment 3 | baseline_bs2000, conv only (the others on the baseline); the same families, conditions (no transform, ramp, reverse ramp, wave), grids (LR x0.6, p x0.8) and starts (the oracle projected onto the family); start and final retrained on held-out seeds 100-107 |
| Search | Global neighbour (noisy_family): each round compares the current schedule with all its neighbours (each parameter one grid step up or down; k = 2 for linear, 4 for power and 2 knots, 6 for 3 knots, 10 for 5 intervals and 5 knots). Every point's score is its mean TTA over full runs on search seeds 0, 1, 2, ... |
| Decision | d = neighbour mean - current mean, SE = 0.12 pp x sqrt(1/n_neighbour + 1/n_current) (the single-run SD of Experiment 6, runs treated as independent). One-sided 80% bounds, Bonferroni over the k neighbours: z = 1.28 / 1.64 / 1.83 / 2.05 for k = 2 / 4 / 6 / 10. Better: d - z SE > 0. Ruled out: d + z SE < 0. Otherwise undecided |
| Moves and seeds | If any neighbour is better, move to the better one with the highest mean. Else run one more seed on the current schedule and on every undecided neighbour, and repeat. Stop when every neighbour is ruled out, or at 8 seeds per point (undecided neighbours then count as not better) |

### *Held-out TTA of the final schedule (Experiment 3's for comparison; the start; search cost)*

| Family | No transform | Ramp | Reverse ramp | Wave |
|---|---|---|---|---|
| 5 constant intervals | **93.80** (Exp. 3: 93.62; start 93.52; 2 moves, 119 runs) | **93.68** (Exp. 3: 93.65; start 93.32; 2 moves, 110 runs) | **93.81** (Exp. 3: 93.86; start 93.81; 0 moves, 76 runs) | **93.60** (Exp. 3: 93.82; start 93.11; 1 moves, 77 runs) |
| Linear decay (1 param) | **93.92** (Exp. 3: 93.86; start 93.92; 0 moves, 15 runs) | **93.42** (Exp. 3: 93.42; start 93.42; 0 moves, 5 runs) | **93.86** (Exp. 3: 93.59; start 93.57; 1 moves, 18 runs) | **93.91** (Exp. 3: 93.91; start 93.72; 1 moves, 8 runs) |
| Power decay (lr, p) | **93.97** (Exp. 3: 93.86; start 93.92; 1 moves, 36 runs) | **93.70** (Exp. 3: 93.95; start 93.42; 1 moves, 21 runs) | **93.86** (Exp. 3: 93.59; start 93.57; 1 moves, 36 runs) | **93.91** (Exp. 3: 93.91; start 93.72; 1 moves, 22 runs) |
| 2 knots (t = 0, 0.5) | **94.03** (Exp. 3: 93.86; start 93.86; 1 moves, 38 runs) | **93.64** (Exp. 3: 93.77; start 93.14; 1 moves, 29 runs) | **93.85** (Exp. 3: 93.88; start 93.85; 0 moves, 34 runs) | **93.79** (Exp. 3: 93.83; start 93.72; 1 moves, 37 runs) |
| 3 knots (t = 0, 1/3, 2/3) | **93.94** (Exp. 3: 93.86; start 93.94; 0 moves, 48 runs) | **93.88** (Exp. 3: 93.68; start 93.74; 1 moves, 55 runs) | **93.90** (Exp. 3: 93.78; start 93.90; 0 moves, 47 runs) | **94.06** (Exp. 3: 94.06; start 93.90; 1 moves, 60 runs) |
| 5 knots (t = 0, 0.2, ..., 0.8) | **93.97** (Exp. 3: 93.94; start 93.97; 0 moves, 79 runs) | **93.91** (Exp. 3: 93.96; start 93.91; 0 moves, 66 runs) | **93.91** (Exp. 3: 93.91; start 93.91; 0 moves, 78 runs) | **93.97** (Exp. 3: 93.94; start 93.97; 0 moves, 84 runs) |

Target: the baseline's 93.92 on the held-out seeds; do-nothing references as in Experiment 3 (ramp 93.42, reverse ramp 93.57, wave 93.72).

### *Final parameter values (seeds behind the final mean; why the search stopped)*

| Family | No transform | Ramp | Reverse ramp | Wave |
|---|---|---|---|---|
| 5 constant intervals | 0.13, 0.13, 0.13, 0.078, 0.017 (8 seeds; max seeds) | 0.6, 0.22, 0.13, 0.028, 0.006 (8 seeds; max seeds) | 0.078, 0.078, 0.13, 0.13, 0.078 (8 seeds; max seeds) | 0.13, 0.078, 0.13, 0.13, 0.028 (8 seeds; max seeds) |
| Linear decay (1 param) | 0.22 (7 seeds; none can be better) | 0.22 (2 seeds; none can be better) | 0.13 (8 seeds; none can be better) | 0.13 (3 seeds; none can be better) |
| Power decay (lr, p) | 0.22, 1.2 (8 seeds; max seeds) | 0.22, 1.2 (7 seeds; none can be better) | 0.13, 1 (8 seeds; max seeds) | 0.13, 1 (5 seeds; none can be better) |
| 2 knots (t = 0, 0.5) | 0.22, 0.078 (8 seeds; none can be better) | 1, 0.078 (8 seeds; max seeds) | 0.047, 0.13 (8 seeds; max seeds) | 0.13, 0.13 (8 seeds; max seeds) |
| 3 knots (t = 0, 1/3, 2/3) | 0.22, 0.13, 0.078 (8 seeds; max seeds) | 1, 0.22, 0.028 (8 seeds; max seeds) | 0.047, 0.078, 0.13 (8 seeds; max seeds) | 0.13, 0.078, 0.13 (8 seeds; max seeds) |
| 5 knots (t = 0, 0.2, ..., 0.8) | 0.22, 0.22, 0.13, 0.078, 0.047 (8 seeds; max seeds) | 1, 0.36, 0.22, 0.078, 0.017 (8 seeds; max seeds) | 0.047, 0.078, 0.078, 0.13, 0.13 (8 seeds; max seeds) | 0.22, 0.078, 0.078, 0.13, 0.078 (8 seeds; max seeds) |

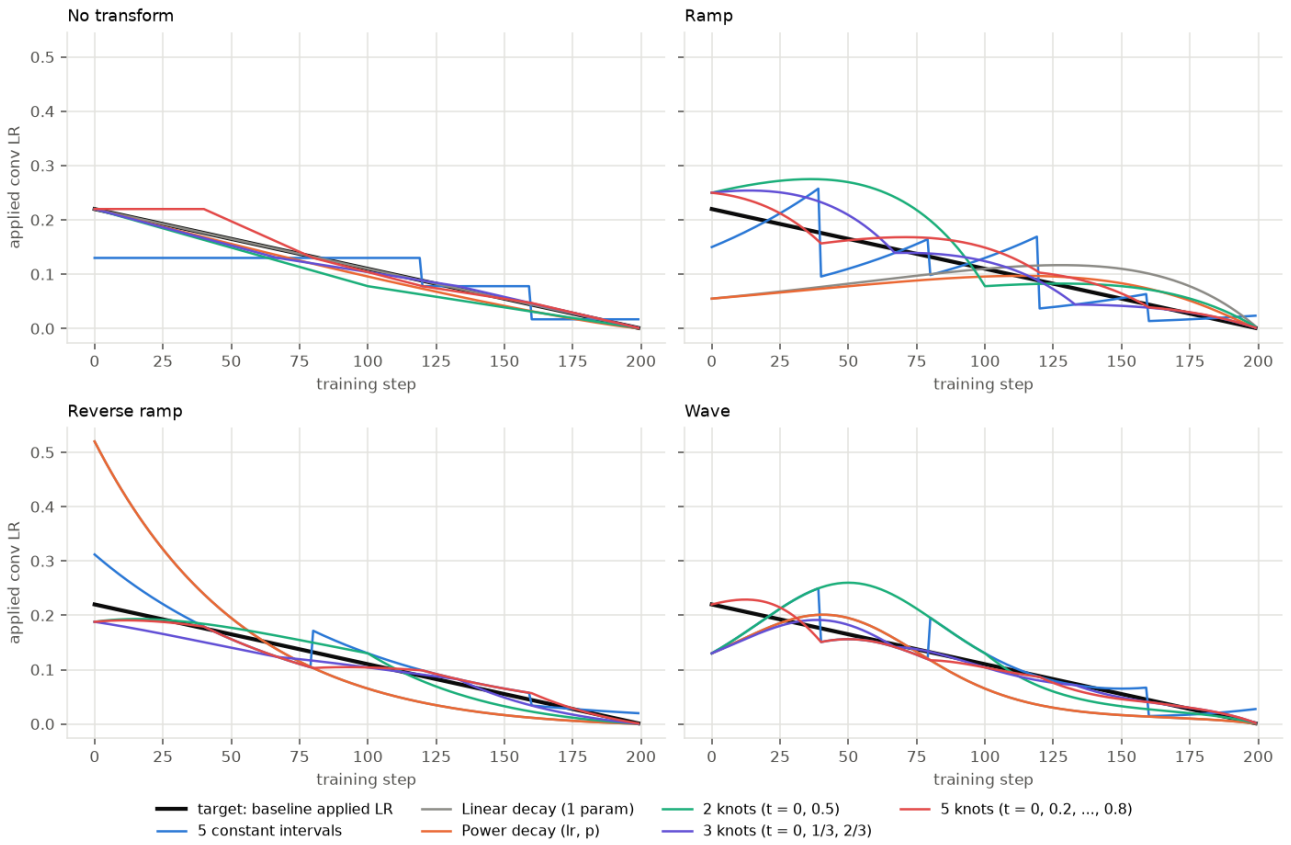



Figure 9. Applied conv LR of each family's final schedule (noise-aware search); the black line is the baseline's applied LR.

### *Observations*

Held-out means over 8 seeds have a standard error of about 0.045 pp, so differences below about 0.1 pp between cells are not resolved. Target: the baseline's 93.92.

The search's own score is now honest. Its final score (the mean over the seeds the search ran it on) is 0.03 pp below the held-out score on average (SD 0.06 over the 24 searches); in Experiment 3 the seed-0 score was 0.15 pp above it (SD 0.15), the winner's curse. Example, 5 constant intervals without a transform: search 93.77, held-out 93.80 (Experiment 3: 93.99 on seed 0, 93.62 held-out).

It does not make the start worse: no final schedule is worse than its start by more than 0.05 pp held-out (mean gain +0.14 pp over the 24 searches); in Experiment 3, 5 of 24 finals were worse than their start by more than 0.05 pp.

It is conservative. It made 0-2 moves per search (Experiment 3: 0-6), and 17 of 24 searches stopped at the 8-seed cap with undecided neighbours: with 8 seeds, a move needs a measured gain of about 0.08-0.12 pp (z = 1.28-2.05 times 0.06 pp), so smaller true gains are left on the table. Against Experiment 3 the held-out results are +0.03 pp

on average: better by more than 0.1 pp in 7 cells (e.g. 5 intervals 93.80 vs 93.62, linear and power under the reverse ramp 93.86 vs 93.59, 2 knots 94.03 vs 93.86), worse in 3 (power under the ramp 93.70 vs 93.95, 5 intervals under the wave 93.60 vs 93.82, 2 knots under the ramp 93.64 vs 93.77), where Experiment 3's longer single-seed walk happened to reach a better region (power, ramp: lr 0.36, p = 2, one and three grid steps from the start).

5 constant intervals without a transform reach 93.80, still below C4's 93.95 (Experiment 5): the search stopped at 0.13, 0.13, 0.13, 0.078, 0.017, three grid steps from C4 (0.13, 0.13, 0.078, 0.047, 0.01), with neighbours still undecided at 8 seeds.

Families: 5 knots is at the target in every condition (93.91-93.97) without moving, as its projected start already was. Linear decay cannot compensate the ramp (93.42, the do-nothing score), as in Experiment 3. The best family per condition: 2 knots 94.03 (no transform), 5 knots 93.91 (ramp, reverse ramp), 3 knots 94.06 (wave).

Cost: 1198 training runs for the 24 searches (Experiment 3: 270), 524 s per search on average (Experiment 3: 114 s), plus 16 held-out runs each.

## Experiment 10. Representation quality of schedule families, guided search

### *Question*

Experiment 3's single-seed search walked far but climbed on seed noise; Experiment 9's significance test removed the noise but rarely moved more than one grid step. Can a search that explores deeper on cheap single runs, pools them with a smooth fit, jumps several steps (and several parameters) at once, and accepts a move only on fresh seeds, get both?

### *Setup*

| Item | Value |
|---|---|
| As Experiment 3 | baseline_bs2000, conv only (the others on the baseline); the same families, conditions, grids (LR x0.6, p x0.8) and starts; start and final retrained on held-out seeds 100-107 |
| 1. Explore | From the current schedule, along each parameter and direction, one grid step at a time, one run each on seed 0, while the run is not clearly worse than the best run on that line (score >= best - 0.84 x 0.12 pp x sqrt(2), one-sided 80%), up to 3 steps |
| 2. Fit | Per parameter, a least-squares quadratic in the grid offset through that line's seed-0 runs (the current schedule's included): the smoothness prior pools 4-7 runs. Predicted gain of an offset = fit there - fit at the current schedule |
| 3. Propose | Each parameter moved alone to its best fitted offset, and the joint move of every parameter with a positive predicted gain (gains summed, a separability prior); the 3 largest predicted gains |
| 4. Verify | On fresh seeds 1-8 (never seed 0, which chose the candidates): better if the lower one-sided 80% bound of (candidate - current) > 0, ruled out if the upper bound < 0, Bonferroni over the candidates (z = 1.50 for 3), SE = 0.12 pp x sqrt(1/n_c + 1/n_current); otherwise one more seed for the current schedule and the undecided candidates, up to 8 |
| Moves | Move to the better candidate with the highest mean and repeat; stop when no parameter is predicted to gain or no candidate proves better |

### *Held-out TTA of the final schedule (Experiments 3 and 9 for comparison; the start; search cost)*

| Family | No transform | Ramp | Reverse ramp | Wave |
|---|---|---|---|---|
| 5 constant intervals | **93.52** (Exp. 3: 93.62; Exp. 9: 93.80; start 93.52; 0 moves, 53 runs) | **93.78** (Exp. 3: 93.65; Exp. 9: 93.68; start 93.32; 2 moves, 91 runs) | **93.73** (Exp. 3: 93.86; Exp. 9: 93.81; start 93.81; 2 moves, 97 runs) | **93.84** (Exp. 3: 93.82; Exp. 9: 93.60; start 93.11; 2 moves, 100 runs) |
| Linear decay (1 param) | **93.92** (Exp. 3: 93.86; Exp. 9: 93.92; start 93.92; 0 moves, 4 runs) | **93.42** (Exp. 3: 93.42; Exp. 9: 93.42; start 93.42; 0 moves, 4 runs) | **93.86** (Exp. 3: 93.59; Exp. 9: 93.86; start 93.57; 1 moves, 7 runs) | **93.91** (Exp. 3: 93.91; Exp. 9: 93.91; start 93.72; 1 moves, 8 runs) |
| Power decay (lr, p) | **93.92** (Exp. 3: 93.86; Exp. 9: 93.97; start 93.92; 0 moves, 9 runs) | **93.88** (Exp. 3: 93.95; Exp. 9: 93.70; start 93.42; 2 moves, 29 runs) | **93.85** (Exp. 3: 93.59; Exp. 9: 93.86; start 93.57; 2 moves, 30 runs) | **93.91** (Exp. 3: 93.91; Exp. 9: 93.91; start 93.72; 1 moves, 15 runs) |
| 2 knots (t = 0, 0.5) | **93.86** (Exp. 3: 93.86; Exp. 9: 94.03; start 93.86; 0 moves, 5 runs) | **93.86** (Exp. 3: 93.77; Exp. 9: 93.64; start 93.14; 2 moves, 20 runs) | **93.85** (Exp. 3: 93.88; Exp. 9: 93.85; start 93.85; 0 moves, 16 runs) | **93.95** (Exp. 3: 93.83; Exp. 9: 93.79; start 93.72; 2 moves, 32 runs) |
| 3 knots (t = 0, 1/3, 2/3) | **93.94** (Exp. 3: 93.86; Exp. 9: 93.94; start 93.94; 0 moves, 22 runs) | **93.88** (Exp. 3: 93.68; Exp. 9: 93.88; start 93.74; 1 moves, 25 runs) | **93.94** (Exp. 3: 93.78; Exp. 9: 93.90; start 93.90; 1 moves, 55 runs) | **94.06** (Exp. 3: 94.06; Exp. 9: 94.06; start 93.90; 1 moves, 34 runs) |
| 5 knots (t = 0, 0.2, ..., 0.8) | **93.97** (Exp. 3: 93.94; Exp. 9: 93.97; start 93.97; 0 moves, 30 runs) | **93.91** (Exp. 3: 93.96; Exp. 9: 93.91; start 93.91; 0 moves, 50 runs) | **93.91** (Exp. 3: 93.91; Exp. 9: 93.91; start 93.91; 0 moves, 11 runs) | **93.97** (Exp. 3: 93.94; Exp. 9: 93.97; start 93.97; 0 moves, 22 runs) |

Target: the baseline's 93.92 on the held-out seeds; do-nothing references as in Experiment 3 (ramp 93.42, reverse ramp 93.57, wave 93.72).

### *Final parameter values (fresh seeds behind the final mean, 'not verified' when the search stopped before testing any candidate; why it stopped)*

| Family | No transform | Ramp | Reverse ramp | Wave |
|---|---|---|---|---|
| 5 constant intervals | 0.22, 0.13, 0.13, 0.078, 0.028 (8 seeds; none proved better) | 0.36, 0.22, 0.078, 0.028, 0.0036 (8 seeds; none proved better) | 0.047, 0.078, 0.13, 0.13, 0.047 (8 seeds; none proved better) | 0.078, 0.078, 0.13, 0.078, 0.01 (8 seeds; none proved better) |
| Linear decay (1 param) | 0.22 (not verified; no predicted gain) | 0.22 (not verified; no predicted gain) | 0.13 (1 seeds; no predicted gain) | 0.13 (2 seeds; no predicted gain) |
| Power decay (lr, p) | 0.22, 1 (not verified; no predicted gain) | 0.36, 1.6 (2 seeds; none proved better) | 0.13, 0.64 (3 seeds; none proved better) | 0.13, 1 (2 seeds; no predicted gain) |
| 2 knots (t = 0, 0.5) | 0.22, 0.13 (not verified; no predicted gain) | 0.6, 0.047 (1 seeds; no predicted gain) | 0.047, 0.13 (3 seeds; none proved better) | 0.017, 0.13 (3 seeds; no predicted gain) |
| 3 knots (t = 0, 1/3, 2/3) | 0.22, 0.13, 0.078 (4 seeds; none proved better) | 1, 0.22, 0.028 (1 seeds; no predicted gain) | 0.047, 0.078, 0.078 (8 seeds; none proved better) | 0.13, 0.078, 0.13 (4 seeds; no predicted gain) |
| 5 knots (t = 0, 0.2, ..., 0.8) | 0.22, 0.22, 0.13, 0.078, 0.047 (5 seeds; none proved better) | 1, 0.36, 0.22, 0.078, 0.017 (8 seeds; none proved better) | 0.047, 0.078, 0.078, 0.13, 0.13 (not verified; no predicted gain) | 0.22, 0.078, 0.078, 0.13, 0.078 (3 seeds; none proved better) |

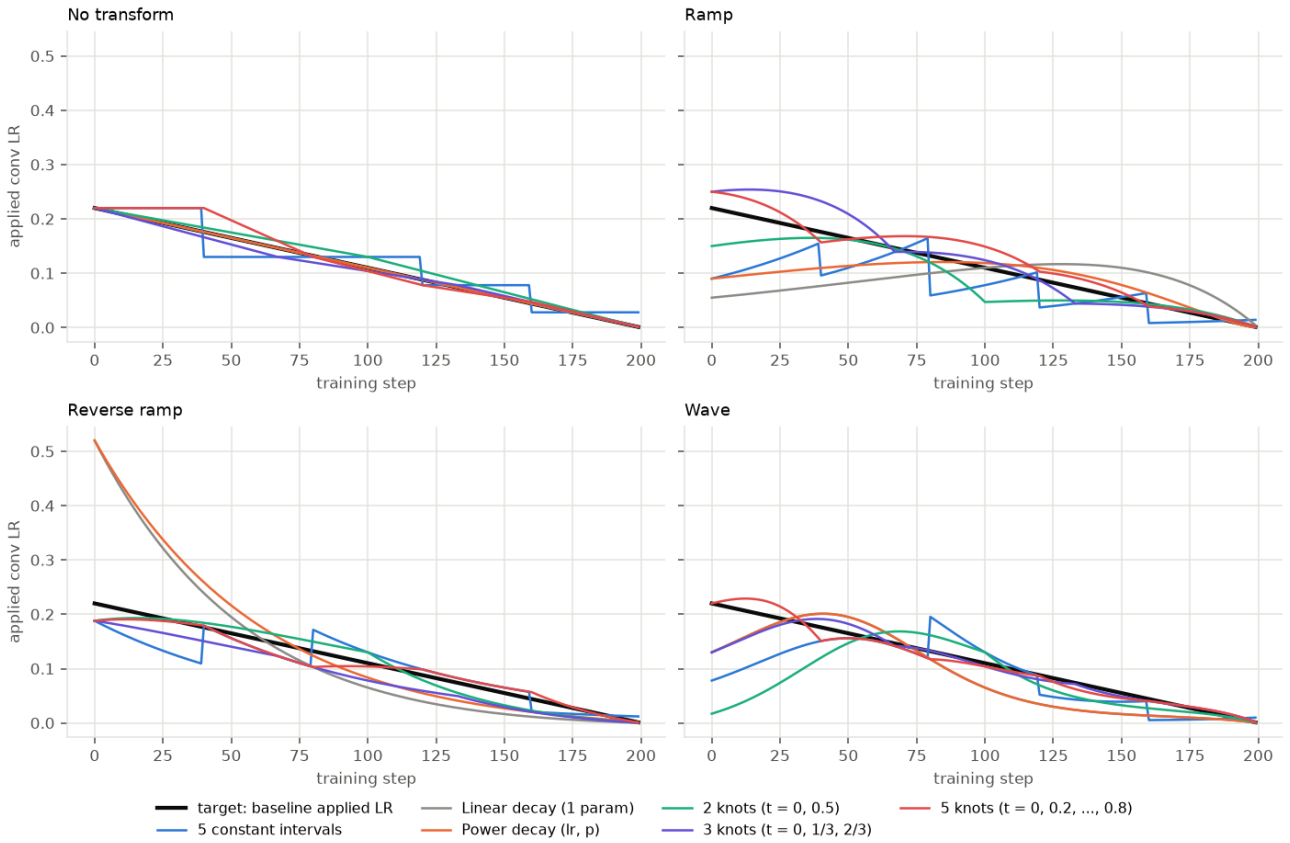



Figure 10. Applied conv LR of each family's final schedule (guided search); the black line is the baseline's applied LR.

### *Observations*

Held-out means over 8 seeds have a standard error of about 0.045 pp, so differences below about 0.1 pp between cells are not resolved. Target: the baseline's 93.92.

Overall the guided search is slightly ahead at lower cost, but not decisively. Mean held-out TTA over the 24 cells: 93.86 (Experiment 9: 93.85, Experiment 3: 93.81); 14 cells within 0.05 pp of the target or above (Experiment 9: 13, Experiment 3: 9). Against Experiment 9 it is better by more than 0.1 pp in 4 cells (5 intervals under the wave 93.84 vs 93.60, power under the ramp 93.88 vs 93.70, 2 knots under the ramp 93.86 vs 93.64 and under the wave 93.95 vs 93.79) and worse in 2 (5 intervals without a transform 93.52 vs 93.80, 2 knots without a transform 93.86 vs 94.03). Cost: 769 training runs for the 24 searches (Experiment 9: 1198, Experiment 3: 270), 337 s per search on average (524 s, 114 s).

Deeper moves happen. Under the ramp, power reached lr 0.36, p = 1.6 (93.88; Experiment 3's lucky walk found lr 0.36, p = 2 at 93.95, Experiment 9 stopped at p = 1.2 with 93.70), and 2 knots reached 0.6, 0.047 (93.86 vs 93.64); 20 accepted moves in total (Experiment 9: 16), several of them multi-step or joint.

The fresh-seed gate keeps the search honest: the final schedule's search score is 0.02 pp above its held-out score on average (SD 0.075, over the 19 searches that verified their final; Experiment 9: -0.03, Experiment 3: +0.15), and only one final is worse than its start by more than 0.05 pp (5 intervals under the reverse ramp, 93.73 vs 93.81).

The weak point is candidate selection. The seed-0 runs along each line carry the same 0.12 pp noise as the effects being measured, so the quadratic fits often follow noise to the end of the explored range. 5 intervals without a transform shows it: the fit sent v0 three steps down to 0.047 on one lucky run (+0.14 pp predicted), the joint move summed four such noisy gains (+0.48 pp predicted), and with v4 to 0.01 these filled all 3 candidate slots; on fresh seeds they measured -0.16, +0.02 and -0.08 pp, so the search stopped at its start (93.52). The first step toward C4, v2 one step down to 0.078, was predicted only +0.05 pp and never tested (v3's line predicted no gain).

Five searches stopped in their first round without testing any candidate (no parameter predicted to gain; 'not verified' in the table); their final is the start.

Likely fixes, not run: rank candidates by predicted gain relative to the fit's own standard error rather than by the raw gain, build the joint move only from parameters whose fitted gain is clearly positive, and test the single-parameter moves before the joint one.

## Experiment 11. Representation quality of schedule families, guided search with uncertainty-aware selection

### *Question*

Experiment 10's weak point was candidate selection: quadratic fits through noisy single runs followed noise to the end of the explored range, and a joint move summing those noisy gains crowded out the real improvements. Does ranking by the fit's lower confidence bound, building the joint move only from clearly positive parameters, and testing single moves first fix it?

### *Setup*

| Item | Value |
|---|---|
| As Experiment 10 | Same families, conditions, starts, exploration (one seed-0 run per point, up to 3 steps per direction), quadratic fit per parameter, fresh-seed verification (seeds 1-8, one-sided 80%, Bonferroni over the candidates raced together) and held-out evaluation |
| Fit uncertainty | The SE of each fitted gain (fit at the offset - fit at the current schedule) from the regression with 0.12 pp noise per run: large where the fit extrapolates or rests on few points |
| Offset choice | Per parameter, the offset (other than the current one) with the highest lower bound, gain - 0.84 SE |
| Candidates | Single-parameter moves with a positive fitted gain, ranked by that lower bound; the best 3 are raced first. Only if none proves better: the joint move of the parameters whose lower bound is above 0 (at least 2), raced alone |

### *Held-out TTA of the final schedule (Experiments 9 and 10 for comparison; the start; search cost)*

| Family | No transform | Ramp | Reverse ramp | Wave |
|---|---|---|---|---|
| 5 constant intervals | **93.52** (Exp. 9: 93.80; Exp. 10: 93.52; start 93.52; 0 moves, 63 runs) | **93.74** (Exp. 9: 93.68; Exp. 10: 93.78; start 93.32; 2 moves, 100 runs) | **93.73** (Exp. 9: 93.81; Exp. 10: 93.73; start 93.81; 2 moves, 94 runs) | **93.66** (Exp. 9: 93.60; Exp. 10: 93.84; start 93.11; 1 moves, 47 runs) |
| Linear decay (1 param) | **93.92** (Exp. 9: 93.92; Exp. 10: 93.92; start 93.92; 0 moves, 4 runs) | **93.42** (Exp. 9: 93.42; Exp. 10: 93.42; start 93.42; 0 moves, 4 runs) | **93.86** (Exp. 9: 93.86; Exp. 10: 93.86; start 93.57; 1 moves, 7 runs) | **93.91** (Exp. 9: 93.91; Exp. 10: 93.91; start 93.72; 1 moves, 8 runs) |
| Power decay (lr, p) | **93.92** (Exp. 9: 93.97; Exp. 10: 93.92; start 93.92; 0 moves, 9 runs) | **93.88** (Exp. 9: 93.70; Exp. 10: 93.88; start 93.42; 2 moves, 29 runs) | **93.85** (Exp. 9: 93.86; Exp. 10: 93.85; start 93.57; 3 moves, 44 runs) | **93.91** (Exp. 9: 93.91; Exp. 10: 93.91; start 93.72; 1 moves, 15 runs) |
| 2 knots (t = 0, 0.5) | **93.86** (Exp. 9: 94.03; Exp. 10: 93.86; start 93.86; 0 moves, 5 runs) | **93.86** (Exp. 9: 93.64; Exp. 10: 93.86; start 93.14; 2 moves, 19 runs) | **93.85** (Exp. 9: 93.85; Exp. 10: 93.85; start 93.85; 0 moves, 16 runs) | **93.93** (Exp. 9: 93.79; Exp. 10: 93.95; start 93.72; 1 moves, 44 runs) |
| 3 knots (t = 0, 1/3, 2/3) | **93.94** (Exp. 9: 93.94; Exp. 10: 93.94; start 93.94; 0 moves, 22 runs) | **93.88** (Exp. 9: 93.88; Exp. 10: 93.88; start 93.74; 1 moves, 25 runs) | **93.94** (Exp. 9: 93.90; Exp. 10: 93.94; start 93.90; 1 moves, 50 runs) | **94.06** (Exp. 9: 94.06; Exp. 10: 94.06; start 93.90; 1 moves, 27 runs) |
| 5 knots (t = 0, 0.2, ..., 0.8) | **93.97** (Exp. 9: 93.97; Exp. 10: 93.97; start 93.97; 0 moves, 30 runs) | **93.93** (Exp. 9: 93.91; Exp. 10: 93.91; start 93.91; 2 moves, 130 runs) | **93.91** (Exp. 9: 93.91; Exp. 10: 93.91; start 93.91; 0 moves, 11 runs) | **93.97** (Exp. 9: 93.97; Exp. 10: 93.97; start 93.97; 0 moves, 22 runs) |

Target: the baseline's 93.92 on the held-out seeds; do-nothing references as in Experiment 3 (ramp 93.42, reverse ramp 93.57, wave 93.72).

***Final parameter values (fresh seeds behind the final mean, 'not verified' when the search stopped before testing any candidate; why it stopped)***

| Family | No transform | Ramp | Reverse ramp | Wave |
|---|---|---|---|---|
| 5 constant intervals | 0.22, 0.13, 0.13, 0.078, 0.028 (8 seeds; none proved better) | 0.6, 0.22, 0.13, 0.028, 0.0036 (8 seeds; none proved better) | 0.047, 0.078, 0.13, 0.13, 0.047 (8 seeds; none proved better) | 0.13, 0.078, 0.13, 0.13, 0.017 (5 seeds; none proved better) |
| Linear decay (1 param) | 0.22 (not verified; no predicted gain) | 0.22 (not verified; no predicted gain) | 0.13 (1 seeds; no predicted gain) | 0.13 (2 seeds; no predicted gain) |
| Power decay (lr, p) | 0.22, 1 (not verified; no predicted gain) | 0.36, 1.6 (2 seeds; none proved better) | 0.13, 0.64 (3 seeds; none proved better) | 0.13, 1 (2 seeds; no predicted gain) |
| 2 knots (t = 0, 0.5) | 0.22, 0.13 (not verified; no predicted gain) | 0.6, 0.047 (1 seeds; no predicted gain) | 0.047, 0.13 (3 seeds; none proved better) | 0.078, 0.13 (8 seeds; none proved better) |
| 3 knots (t = 0, 1/3, 2/3) | 0.22, 0.13, 0.078 (4 seeds; none proved better) | 1, 0.22, 0.028 (1 seeds; no predicted gain) | 0.047, 0.078, 0.078 (8 seeds; none proved better) | 0.13, 0.078, 0.13 (3 seeds; no predicted gain) |
| 5 knots (t = 0, 0.2, ..., 0.8) | 0.22, 0.22, 0.13, 0.078, 0.047 (5 seeds; none proved better) | 0.6, 0.36, 0.22, 0.047, 0.01 (8 seeds; none proved better) | 0.047, 0.078, 0.078, 0.13, 0.13 (not verified; no predicted gain) | 0.22, 0.078, 0.078, 0.13, 0.078 (3 seeds; none proved better) |

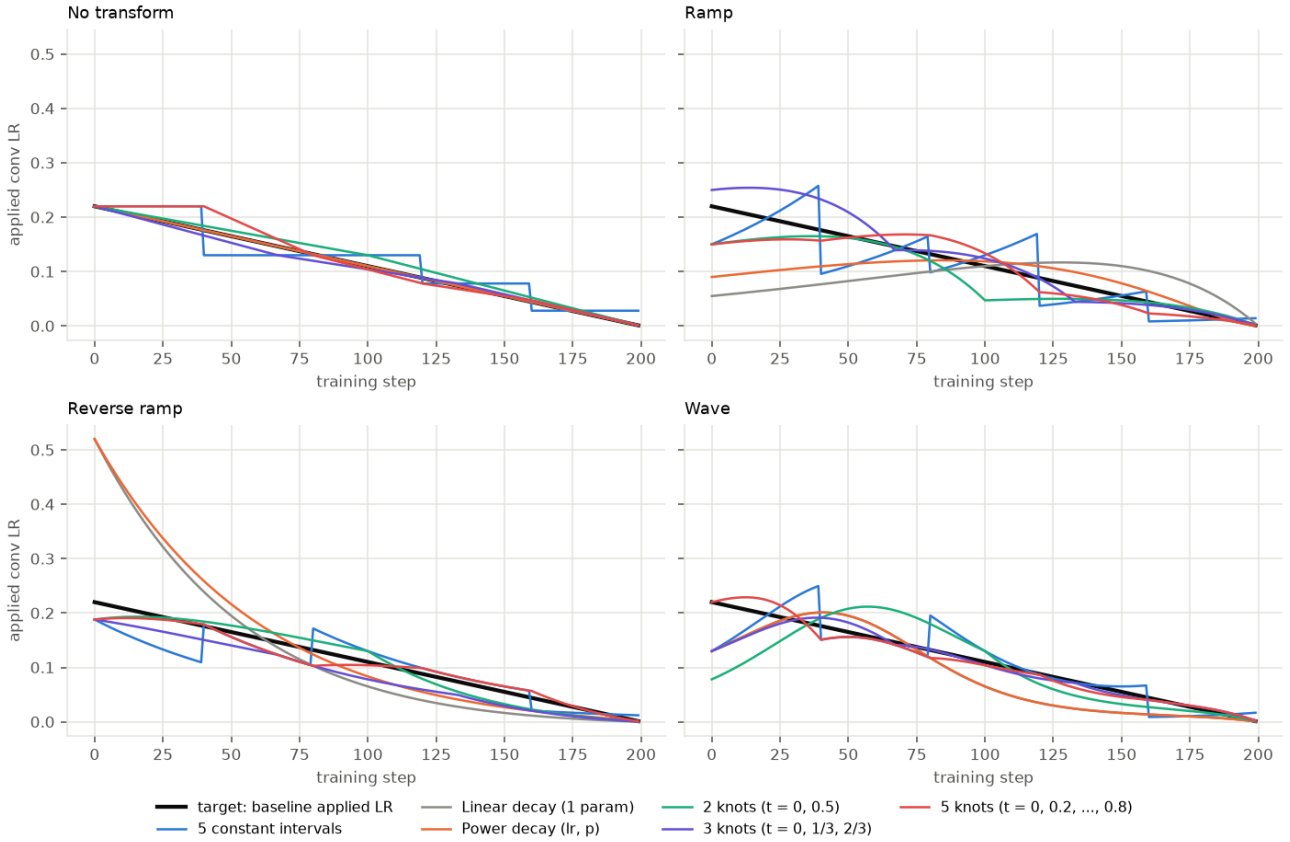



Figure 11. Applied conv LR of each family's final schedule (guided search, uncertainty-aware selection); the black line is the baseline's applied LR.

## Experiment 12. Which lookahead decides constant intervals like the final result, and how noisy is it?

### *Question*

An interval search commits each constant interval by scoring candidates with a lookahead from the same trained state. Which lookahead ranks the candidates like their true effect on the final score, and how noisy is one seed's comparison of two candidates scored from the same state?

### *Setup*

| Item | Value |
|---|---|
| Plans | conv as 5 constant intervals, the other groups on the baseline. BS 2000: C4 (0.13, 0.13, 0.078, 0.047, 0.01; matches the baseline, Experiment 5); BS 500 and 125: the q = 0.75 staircases of Experiment 16 (0.11, 0.085, 0.058, 0.033, 0.0065 and 0.04, 0.031, 0.021, 0.012, 0.0024) |
| Candidates | At interval k, from the plan's trained state: the plan's value moved -2..+2 grid steps (x0.6), trained for the interval |
| Scores | truth: the rest of the run on the plan (what the decision affects); rest_lin: the rest of the run with conv decaying linearly from the candidate to 0; c40_m: one interval (a fifth of the run) at m x candidate (m = 0.6, 0.36, 0.22); l40: one interval decaying linearly to 0 |
| Seeds | BS 2000 and 500: 8 (0-7), BS 125: 4; each seed is its own run from initialization |
| Decision quality | 1-seed hit: how often one seed's best candidate by the score is the candidate with the best mean truth; regret: the mean, over seeds, of the mean truth lost by that pick (pp) |

### *BS 2000 (8 seeds): mean pick; 1-seed hit rate; 1-seed regret (pp of final TTA)*

| Interval (best by truth) | truth | rest_lin | c40_0.22 | c40_0.36 | c40_0.6 | l40 |
|---|---|---|---|---|---|---|
| 1 (0.22) | 0.22; 0.25; 0.039 | 0.13; 0.38; 0.001 | 0.22; 0.38; 0.001 | 0.13; 0.00; 0.001 | 0.078; 0.00; 0.032 | 0.13; 0.00; 0.001 |
| 2 (0.13) | 0.13; 0.50; 0.028 | 0.13; 0.88; 0.011 | 0.078; 0.38; 0.082 | 0.078; 0.00; 0.115 | 0.047; 0.00; 0.294 | 0.078; 0.12; 0.079 |
| 3 (0.078) | 0.078; 0.62; 0.009 | 0.078; 0.62; 0.017 | 0.078; 0.50; 0.046 | 0.047; 0.12; 0.080 | 0.028; 0.00; 0.134 | 0.047; 0.25; 0.068 |
| 4 (0.047) | 0.047; 0.50; 0.029 | 0.047; 0.25; 0.050 | 0.047; 0.50; 0.029 | 0.028; 0.25; 0.037 | 0.028; 0.00; 0.083 | 0.047; 0.25; 0.050 |
| 5 (0.01) | 0.01; 0.50; 0.027 | - | - | - | - | - |

### *BS 500 (8 seeds): mean pick; 1-seed hit rate; 1-seed regret (pp of final TTA)*

| Interval (best by truth) | truth | rest_lin | c40_0.22 | c40_0.36 | c40_0.6 | l40 |
|---|---|---|---|---|---|---|
| 1 (0.078) | 0.078; 0.38; 0.027 | 0.078; 0.88; 0.006 | 0.078; 0.88; 0.006 | 0.078; 1.00; 0.000 | 0.047; 0.25; 0.035 | 0.078; 1.00; 0.000 |
| 2 (0.078) | 0.078; 0.25; 0.016 | 0.047; 0.38; 0.013 | 0.047; 0.00; 0.021 | 0.047; 0.00; 0.074 | 0.028; 0.00; 0.126 | 0.047; 0.00; 0.021 |
| 3 (0.028) | 0.028; 0.38; 0.019 | 0.028; 0.25; 0.008 | 0.028; 0.88; 0.012 | 0.028; 0.62; 0.024 | 0.017; 0.25; 0.071 | 0.028; 0.75; 0.001 |
| 4 (0.028) | 0.028; 0.75; 0.029 | 0.028; 0.50; 0.059 | 0.028; 0.75; 0.019 | 0.017; 0.50; 0.039 | 0.017; 0.12; 0.068 | 0.028; 0.50; 0.059 |
| 5 (0.006) | 0.006; 0.25; 0.018 | - | - | - | - | - |

### *BS 125 (4 seeds): mean pick; 1-seed hit rate; 1-seed regret (pp of final TTA)*

| Interval (best by truth) | truth | rest_lin | c40_0.22 | c40_0.36 | c40_0.6 | l40 |
|---|---|---|---|---|---|---|
| 1 (0.028) | 0.028; 0.50; 0.017 | 0.028; 1.00; 0.000 | 0.028; 0.75; 0.009 | 0.028; 1.00; 0.000 | 0.028; 0.75; 0.023 | 0.028; 1.00; 0.000 |
| 2 (0.047) | 0.047; 0.50; 0.005 | 0.028; 0.00; 0.051 | 0.017; 0.00; 0.132 | 0.017; 0.00; 0.167 | 0.01; 0.00; 0.163 | 0.017; 0.00; 0.132 |
| 3 (0.017) | 0.017; 0.50; 0.036 | 0.017; 0.75; 0.018 | 0.01; 0.00; 0.072 | 0.01; 0.00; 0.072 | 0.01; 0.00; 0.089 | 0.01; 0.00; 0.081 |
| 4 (0.01) | 0.01; 0.75; 0.017 | 0.01; 0.75; 0.051 | 0.01; 0.50; 0.069 | 0.006; 0.25; 0.052 | 0.006; 0.00; 0.175 | 0.01; 0.75; 0.051 |
| 5 (0.0036) | 0.0036; 0.50; 0.039 | - | - | - | - | - |

### *Noise of one seed's comparison of two candidates*

| Interval | Noise of a paired comparison at BS 2000 (SD over seeds of candidate - plan value, pp): truth / rest_lin |
|---|---|
| 1 | 0.047: 0.23 / 0.17; 0.078: 0.27 / 0.22; 0.22: 0.24 / 0.15; 0.36: 0.25 / 0.19 |
| 2 | 0.047: 0.20 / 0.17; 0.078: 0.12 / 0.16; 0.22: 0.21 / 0.14; 0.36: 0.16 / 0.25 |
| 3 | 0.028: 0.16 / 0.19; 0.047: 0.14 / 0.19; 0.13: 0.15 / 0.22; 0.22: 0.20 / 0.16 |
| 4 | 0.017: 0.10 / 0.09; 0.028: 0.10 / 0.07; 0.078: 0.17 / 0.13; 0.13: 0.15 / 0.19 |
| 5 | 0.0036: 0.07 / -; 0.006: 0.06 / -; 0.017: 0.10 / -; 0.028: 0.13 / - |

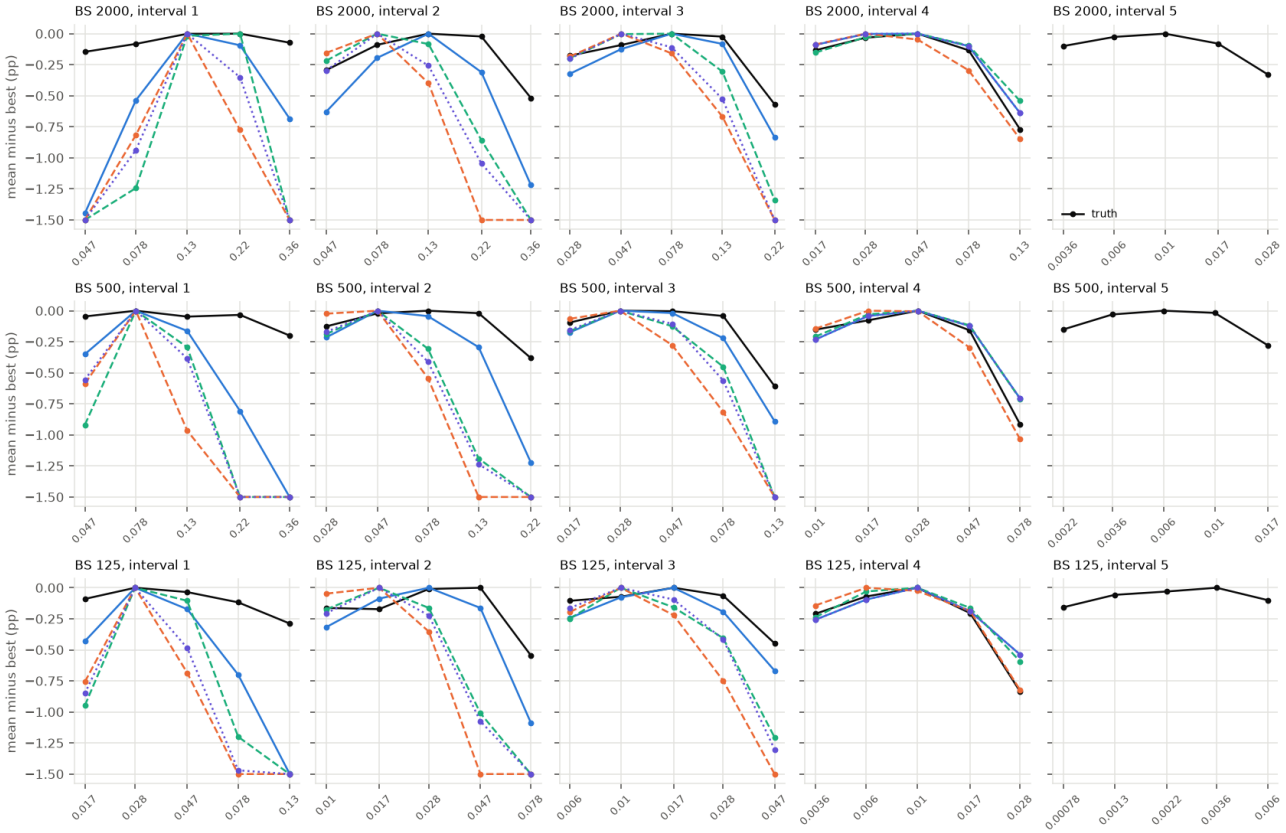



Figure 12. Each score's mean per candidate over seeds, relative to its own best (points below -1.5 drawn at -1.5). A lookahead decides like the truth when its curve peaks where the truth's does.

### *Observations*

The rest-of-run linear lookahead decides like the truth: its 8-seed pick is the truth's best in intervals 2-4 (interval 5 has no lookahead), and in interval 1 it picks 0.13 where the truth ties 0.13 and 0.22 (93.83 both). One seed's pick by it loses 0.001-0.050 pp of final TTA per interval, about what one seed's pick by the truth itself loses (0.009-0.039): the truth is flat near its optimum, so a neighbouring pick costs little.

One-interval (40-step) lookaheads are biased toward small LRs once training is under way: in interval 2 every one of them picks 0.047-0.078 where the truth peaks at 0.13 (regret 0.08-0.29 pp per seed), and c40_0.6 picks 1-2 grid steps below the truth's best in intervals 2-4. The mildest cooldown, c40_0.22, is the least biased (regret 0.03-0.08 pp from interval 2 on) but still loses to rest_lin in intervals 2 and 3.

Scoring candidates from the same trained state does not cancel the seed noise: one seed's difference between two candidates' truths has an SD of 0.12-0.27 pp in intervals 1-2, as large as two independent full runs (0.17), falling to 0.06-0.17 pp in intervals 4-5 as fewer steps remain. Differences of 0.05-0.1 pp between neighbouring candidates are therefore not resolved by one seed, but they also cost little.

Cost at BS 2000: rest_lin costs the rest of the run per candidate (200, 160, 120, 80, 40 steps for intervals 1-5), so about 3 candidates per interval cost 1800 steps = 9 x a run for one searched LR, within the campaign's 20x budget.

At BS 500 (8 seeds) and BS 125 (4 seeds) the picture is the same, with flatter optima. The truth's top two or three candidates in each interval are within about 0.05 pp of each other, so one seed's pick by any good score costs little (rest_lin: 0.006-0.059 pp per interval at BS 500, 0.000-0.051 at BS 125). The short cooldowns again pick low where training is under way: at BS 125, interval 2, they pick 0.01-0.017 where the truth peaks at 0.047 (regret

0.13-0.17 pp), and in interval 3 they pick 0.01 for the truth's 0.017; at BS 500 the 0.22 x cooldown is one step low in interval 2 only (0.047 for 0.078). Over all three batch sizes, one seed's decision costs on average 0.020 pp with rest_lin and 0.036 pp with c40_0.22; picking the largest candidate within 0.1-0.3 pp of the best instead does not reduce either (noise, not bias, dominates single decisions).

## Experiment 13. ICS v1: constant intervals with a rest-of-run lookahead, all batch sizes and transforms

### *Question*

Does the interval-cooldown search, with the lookahead Experiment 12 found faithful, match the baseline at batch sizes 2000, 500 and 125, with and without transforms, searching conv only or all four LRs, within 20 full runs?

### *Setup*

| Item | Value |
|---|---|
| Representation | Each searched LR: 5 constant intervals (spec values; the transform applies on top), on the x0.6 grid |
| Per interval | Climb the lead LR (conv) from the previous interval's value, smaller values first, moving while the score improves. A candidate's score: train the interval from the committed state, then the rest of the run with every searched LR decaying linearly from its candidate value to 0 (rest_lin); the last interval is scored by the final TTA. Commit the best and continue from its trained state |
| Conv only (a) | conv from its baseline initial LR; head, norm_bias, whiten_bias on the baseline |
| 4 LRs (b) | All four from 1.0. Interval 1: each LR's level is scanned on a x7.7 grid (both neighbours first, then the better side while improving), scored after the interval plus one interval at 0.36 x the value; then conv is refined on the x0.6 grid with rest_lin. Later intervals: conv is climbed and the other three follow its profile (value_k = value_1 x conv_k / conv_1) |
| Budget | At most 20 full runs of train steps (every trained step counts); the transform applies to the searched groups |
| Evaluation | 2 search seeds (0, 1) per cell; each committed schedule retrained on held-out seeds 100-107 |

### *Held-out TTA (mean over search seeds; vs the baseline; per search seed); cost in full runs*

| Batch size, scenario (baseline) | none | ramp | reverse ramp | wave |
|---|---|---|---|---|
| BS 2000, conv only (93.92) | **93.82** (-0.10; 93.82/93.82); 10.0x | **93.15** (-0.77; 93.19/93.11); 10.6x; do-nothing 93.42 | **93.80** (-0.12; 93.81/93.79); 11.0x; do-nothing 93.57 | **93.54** (-0.38; 93.61/93.46); 9.6x; do-nothing 93.72 |
| BS 2000, 4 LRs (93.92) | **92.85** (-1.07; 93.68/92.01); 17.2x | **91.65** (-2.27; 90.82/92.48); 15.6x; do-nothing 93.25 | **93.70** (-0.22; 93.89/93.51); 15.2x; do-nothing 93.45 | **92.17** (-1.75; 91.94/92.40); 15.6x; do-nothing 93.68 |
| BS 500, conv only (94.00) | **93.84** (-0.16; 93.75/93.92); 9.4x | **93.40** (-0.60; 93.27/93.53); 10.4x; do-nothing 93.37 | **93.88** (-0.11; 93.95/93.82); 11.8x; do-nothing 93.68 | **93.50** (-0.50; 93.25/93.75); 10.4x; do-nothing 93.77 |
| BS 500, 4 LRs (94.00) | **93.58** (-0.42; 93.53/93.62); 15.8x | **92.93** (-1.07; 92.79/93.07); 16.6x; do-nothing 93.32 | running | running |
| BS 125, conv only (93.86) | **93.73** (-0.13; 93.70/93.76); 9.6x | running | **93.66** (-0.20; 93.72/93.60); 10.2x; do-nothing 93.68 | running |
| BS 125, 4 LRs (93.86) | running | running | running | running |

Cells within 0.1 pp of the baseline (about 2 SE of an 8-seed mean) or above: 1 of 16 finished. * = the search hit its step budget. Do-nothing: the baseline schedule kept under the transform.

### *Committed schedules (search seed 0)*

| Cell | Committed schedules (search seed 0) |
|---|---|
| BS 2000, conv only, none | conv: 0.13, 0.13, 0.13, 0.047, 0.006 |
| BS 2000, conv only, ramp | conv: 0.13, 0.13, 0.047, 0.017, 0.0036 |
| BS 2000, conv only, reverse ramp | conv: 0.078, 0.13, 0.078, 0.078, 0.028 |
| BS 2000, conv only, wave | conv: 0.22, 0.13, 0.13, 0.078, 0.01 |
| BS 2000, 4 LRs, none | conv: 0.13, 0.13, 0.047, 0.028, 0.006; head: 3500, 3500, 1300, 750, 160; norm_bias: 0.13, 0.13, 0.047, 0.028, 0.006; whiten_bias: 460, 460, 170, 99, 21 |
| BS 2000, 4 LRs, ramp | conv: 0.13, 0.13, 0.047, 0.017, 0.0022; head: 1, 1, 0.36, 0.13, 0.017; norm_bias: 1, 1, 0.36, 0.13, 0.017; whiten_bias: 0.13, 0.13, 0.047, 0.017, 0.0022 |
| BS 2000, 4 LRs, reverse ramp | conv: 0.078, 0.078, 0.078, 0.078, 0.028; head: 460, 460, 460, 460, 170; norm_bias: 60, 60, 60, 60, 22; whiten_bias: 1, 1, 1, 1, 0.36 |
| BS 2000, 4 LRs, wave | conv: 0.13, 0.13, 0.078, 0.078, 0.01; head: 1, 1, 0.6, 0.6, 0.077; norm_bias: 1, 1, 0.6, 0.6, 0.077; whiten_bias: 1, 1, 0.6, 0.6, 0.077 |
| BS 500, conv only, none | conv: 0.078, 0.078, 0.047, 0.047, 0.01 |
| BS 500, conv only, ramp | conv: 0.078, 0.078, 0.028, 0.01, 0.0022 |
| BS 500, conv only, reverse ramp | conv: 0.028, 0.047, 0.047, 0.028, 0.017 |
| BS 500, conv only, wave | conv: 0.13, 0.13, 0.078, 0.047, 0.01 |
| BS 500, 4 LRs, none | conv: 0.078, 0.047, 0.047, 0.028, 0.006; head: 460, 280, 280, 170, 35; norm_bias: 0.017, 0.01, 0.01, 0.0061, 0.0013; whiten_bias: 7.7, 4.6, 4.6, 2.8, 0.59 |
| BS 500, 4 LRs, ramp | conv: 0.078, 0.047, 0.028, 0.01, 0.0013; head: 3500, 2100, 1300, 450, 58; norm_bias: 7.7, 4.6, 2.8, 0.99, 0.13; whiten_bias: 1, 0.6, 0.36, 0.13, 0.017 |
| BS 125, conv only, none | conv: 0.028, 0.028, 0.017, 0.006, 0.0036 |
| BS 125, conv only, reverse ramp | conv: 0.017, 0.01, 0.01, 0.01, 0.0036 |

### *Observations*

Targets (held-out seeds 100-107): baselines 93.92 (BS 2000), 94.00 and 93.86 (BS 500 and 125, from the post-campaign reruns; the table's own baseline runs replace these once they finish). One held-out mean has a standard error of about 0.045 pp, so differences below about 0.1 pp are not resolved.

Conv only without a transform, v1 lands 0.1-0.2 pp short at every batch size: 93.82 (both search seeds) at BS 2000, 93.84 at BS 500 and 93.73 at BS 125, for 9.4-10 full runs of train steps. Its BS 2000 schedules (0.13, 0.13, 0.13, 0.047, 0.006 and 0.22, 0.13, 0.078, 0.047, 0.01) are one or two grid steps from C4 (93.95).

Under transforms the rest-of-run linear lookahead fails: the ramp gives 93.15 (BS 2000) and 93.40 (BS 500), below the do-nothing 93.42 / 93.37, and the wave 93.54 / 93.50. The lookahead decays linearly in the spec schedule, so under a transform its applied shape is distorted: under the ramp the applied LR rises late in the lookahead, so high early candidates look bad and the early intervals stay at 0.13-0.22 where the compensating spec is about 0.9. The reverse ramp, which mostly needs smaller early values, is fine (93.80 / 93.88).

From scratch with all four LRs (b), the level scan fails: 91.65-93.70 at BS 2000 and 92.93-93.58 at BS 500. Scored after one interval, head and norm_bias look flat (87.3% at levels 0.13, 1 and 7.7 in one search), so the scan leaves them near their start of 1.0, where a full run loses about 1 pp for head (Experiment 7). The search seed 1 run without a transform (92.01) kept head and norm_bias at 1.0.

The search's own final score is optimistic by up to 0.2 pp at BS 500 and 125 (e.g. 94.06 own vs 93.92 held-out; 93.94 vs 93.76): its decisions are selected on one seed.

v1 was stopped after these cells (the remaining transformed and 4-LR cells were cancelled) in favour of v2.

## Experiment 14. ICS v2: a short-cooldown pass, then a rest-of-run pass on its plan

### *Question*

v1's rest-of-run linear lookahead is defined on the spec schedule, so under a transform its applied shape is distorted and it misjudges early intervals (v1 under the ramp: 93.11-93.19, below doing nothing). Does a first pass with a short, local cooldown, whose schedule then serves as the second pass's lookahead, fix that?

### *Setup*

| Item | Value |
|---|---|
| Pass 1 | Every interval in order: climb conv from the previous interval's value, each candidate scored after the interval plus one interval at 0.22 x its value (the least biased short cooldown, Experiment 12; local, so a transform's global shape does not distort it) |
| Pass 2 | From the initial state again: each interval's climb starts at pass 1's value; a candidate is scored after the rest of the run on pass 1's schedule, its tail scaled by candidate / pass-1 value (rest_plan); the last interval by the final TTA |
| Other | As v1 (5 constant intervals, x0.6 grid, budget 20 full runs, 2 search seeds, held-out seeds 100-107); conv only: v2's 4-LR variant is replaced by v3/v4 (level rule) |

### *Held-out TTA (mean over search seeds; vs the baseline; per search seed); cost in full runs*

| Batch size, scenario (baseline) | none | ramp | reverse ramp | wave |
|---|---|---|---|---|
| BS 2000, conv only (93.92) | **93.70** (-0.22; 93.77/93.62); 16.2x | **93.72** (-0.20; 93.68/93.77); 19.8x; do-nothing 93.42 | **93.77** (-0.15; 93.83/93.71); 16.6x; do-nothing 93.57 | **93.68** (-0.24; 93.84/93.53); 16.0x; do-nothing 93.72 |
| BS 2000, 4 LRs (93.92) | running | running | running | running |
| BS 500, conv only (94.00) | **93.88** (-0.11; 93.89/93.88); 16.6x | **93.61** (-0.39; 93.54/93.68); 20.2x*; do-nothing 93.37 | **93.81** (-0.19; 93.81/93.81); 16.6x; do-nothing 93.68 | **93.78** (-0.22; 93.70/93.85); 15.6x; do-nothing 93.77 |
| BS 500, 4 LRs (94.00) | running | running | running | running |
| BS 125, conv only (93.86) | **93.47** (-0.39; 93.43/93.52); 16.0x | **93.37** (-0.49; 93.50/93.25); 18.0x; do-nothing 93.48 | **93.75** (-0.11; 93.72/93.78); 18.8x; do-nothing 93.68 | **93.55** (-0.31; 93.40/93.70); 15.6x; do-nothing 93.69 |
| BS 125, 4 LRs (93.86) | running | running | running | running |

Cells within 0.1 pp of the baseline (about 2 SE of an 8-seed mean) or above: 0 of 12 finished. * = the search hit its step budget. Do-nothing: the baseline schedule kept under the transform.

### *Committed schedules (search seed 0)*

| Cell | Committed schedules (search seed 0) |
|---|---|
| BS 2000, conv only, none | conv: 0.13, 0.13, 0.078, 0.078, 0.017 |
| BS 2000, conv only, ramp | conv: 0.36, 0.22, 0.047, 0.017, 0.0036 |
| BS 2000, conv only, reverse ramp | conv: 0.078, 0.078, 0.078, 0.028, 0.017 |
| BS 2000, conv only, wave | conv: 0.078, 0.078, 0.078, 0.047, 0.01 |
| BS 500, conv only, none | conv: 0.078, 0.047, 0.047, 0.017, 0.0036 |
| BS 500, conv only, ramp | conv: 0.22, 0.078, 0.017, 0.017, 0.0022 |
| BS 500, conv only, reverse ramp | conv: 0.047, 0.047, 0.028, 0.028, 0.017 |
| BS 500, conv only, wave | conv: 0.047, 0.028, 0.017, 0.028, 0.006 |
| BS 125, conv only, none | conv: 0.028, 0.017, 0.006, 0.01, 0.0022 |
| BS 125, conv only, ramp | conv: 0.13, 0.028, 0.006, 0.0036, 0.00047 |
| BS 125, conv only, reverse ramp | conv: 0.028, 0.017, 0.028, 0.017, 0.006 |
| BS 125, conv only, wave | conv: 0.017, 0.006, 0.01, 0.006, 0.0036 |

### *Observations*

The short-cooldown first pass fixes v1's transform failure: under the ramp the held-out TTA rises from 93.15 (v1) to 93.72 at BS 2000 and from 93.40 to 93.61 at BS 500, because pass 1's local cooldown lets the early intervals rise (0.36-0.6 at BS 2000, where v1 stayed at 0.13-0.22) and pass 2 then judges candidates against a lookahead with the right, transform-shaped profile.

It is still 0.1-0.5 pp short everywhere: 93.68-93.77 at BS 2000 (baseline 93.92), 93.61-93.88 at BS 500 (94.00) and 93.37-93.75 at BS 125 (93.86), for 15-20 full runs. At BS 125 it is worse than v1 without a transform (93.47 vs 93.73): pass 1 decays faster than v1's rest-of-run lookahead (0.028, 0.017, 0.006, ... vs 0.028, 0.028, 0.017, ...) and pass 2, starting each climb from pass 1's values, does not undo it.

Search-seed spread is large (up to 0.31 pp between the two seeds of one cell), so single-seed decisions still dominate the error (Experiment 17).

## Experiment 15. ICS v5: v2 plus kept first interval and replicated late decisions

### *Question*

The late intervals' decisions are worth ~0.1 pp each and one seed gets them wrong about half the time (Experiment 12), while the first interval's rest-of-run candidates cost a full run each. Does keeping pass 1's first interval and scoring the late candidates by several continuations close the remaining gap within budget?

### *Setup*

| Item | Value |
|---|---|
| As v2 | Pass 1 with the 0.22 x short cooldown, pass 2 with the rest_plan lookahead |
| Kept interval | Pass 2 keeps pass 1's first interval (value and trained state, at no cost) |
| Replicates | In pass 2, a candidate for interval 4 is scored by the mean of 2 continuations and for interval 5 of 3: the first on the run's own data stream (its state is committed), the others restore the pre-interval state, reseed the data order and augmentation (fresh epoch) and train interval + lookahead again |
| 4 LRs | Levels: x7.7 scan right after interval 1 (conv: best score; head and biases: largest within 1 pp of the best), then as conv only with the other groups tied to conv's profile |

### *Held-out TTA (mean over search seeds; vs the baseline; per search seed); cost in full runs*

| Batch size, scenario (baseline) | none | ramp | reverse ramp | wave |
|---|---|---|---|---|
| BS 2000, conv only (93.92) | **93.75** (-0.17; 93.95/93.56); 14.6x | **93.58** (-0.34; 93.66/93.51); 17.0x; do-nothing 93.42 | **93.81** (-0.11; 93.91/93.71); 18.6x; do-nothing 93.57 | **93.79** (-0.13; 93.84/93.75); 15.4x; do-nothing 93.72 |
| BS 2000, 4 LRs (93.92) | running | running | running | running |
| BS 500, conv only (94.00) | **93.86** (-0.14; 93.83/93.89); 16.2x | **93.65** (-0.35; 93.62/93.68); 19.6x*; do-nothing 93.37 | **93.81** (-0.19; 93.81/93.81); 16.0x; do-nothing 93.68 | **93.76** (-0.24; 93.68/93.84); 15.0x; do-nothing 93.77 |
| BS 500, 4 LRs (94.00) | running | running | running | running |
| BS 125, conv only (93.86) | **93.53** (-0.33; 93.45/93.61); 16.8x | **93.42** (-0.44; 93.40/93.43); 16.4x; do-nothing 93.48 | **93.72** (-0.14; 93.66/93.79); 14.8x; do-nothing 93.68 | **93.61** (-0.25; 93.52/93.70); 15.6x; do-nothing 93.69 |
| BS 125, 4 LRs (93.86) | running | running | running | running |

Cells within 0.1 pp of the baseline (about 2 SE of an 8-seed mean) or above: 0 of 12 finished. * = the search hit its step budget. Do-nothing: the baseline schedule kept under the transform.

### *Committed schedules (search seed 0)*

| Cell | Committed schedules (search seed 0) |
|---|---|
| BS 2000, conv only, none | conv: 0.13, 0.13, 0.078, 0.047, 0.01 |
| BS 2000, conv only, ramp | conv: 0.36, 0.22, 0.047, 0.01, 0.0022 |
| BS 2000, conv only, reverse ramp | conv: 0.078, 0.078, 0.078, 0.078, 0.028 |
| BS 2000, conv only, wave | conv: 0.078, 0.078, 0.078, 0.047, 0.006 |
| BS 500, conv only, none | conv: 0.078, 0.047, 0.047, 0.028, 0.0036 |
| BS 500, conv only, ramp | conv: 0.22, 0.078, 0.017, 0.01, 0.0013 |
| BS 500, conv only, reverse ramp | conv: 0.047, 0.047, 0.028, 0.028, 0.017 |
| BS 500, conv only, wave | conv: 0.047, 0.028, 0.017, 0.017, 0.006 |
| BS 125, conv only, none | conv: 0.028, 0.017, 0.006, 0.0036, 0.0013 |
| BS 125, conv only, ramp | conv: 0.078, 0.017, 0.01, 0.0036, 0.00047 |
| BS 125, conv only, reverse ramp | conv: 0.017, 0.01, 0.01, 0.006, 0.006 |
| BS 125, conv only, wave | conv: 0.017, 0.006, 0.01, 0.006, 0.0013 |

### *Observations*

v5 is v2 made cheaper at the start and less noisy at the end, and it is no better on average: 93.58-93.81 at BS 2000 (baseline 93.92), 93.65-93.86 at BS 500 (94.00) and 93.42-93.72 at BS 125 (93.86), for 14-20 full runs. Its best single result is the C4 schedule itself (0.13, 0.13, 0.078, 0.047, 0.01; 93.95 at BS 2000, search seed 0), so the method can find a matching schedule but does not reliably.

At BS 500 and 125 it is systematically too low: its BS 125 schedules (e.g. 0.028, 0.017, 0.006, 0.0036, 0.0013) sit one to two grid steps below the 5-interval staircase that scores 93.80 (0.04, 0.031, 0.021, 0.012, 0.0024), and its BS 500 schedules start at 0.078 where the staircase starts at 0.11. The first pass's short cooldown picks low, and the second pass does not raise it, probably because its candidates scale the whole remaining plan, so raising a middle interval also raises the end of the run, which was already right (v10 tests this by keeping the tail unscaled).

Replicating the last two intervals' decisions did not show a measurable gain over v2 (same cells, 2 search seeds each), which is consistent with the late decisions costing little either way (Experiment 12).

## Experiment 16. How well do constant intervals represent the baseline?

### *Question*

Before asking a search to find a constant-interval schedule, can such a schedule match the baseline at all, at each batch size and under each transform, with 5 or 10 intervals?

### *Setup*

| Item | Value |
|---|---|
| Schedules | conv's schedule replaced by n equal constant intervals, each at the oracle spec (the baseline's applied LR divided by the transform) a fraction q of the way through the interval; the last interval at min(q, 0.95) (the oracle reaches 0 at the end). q = 0.5 is about the interval mean, q = 1 the end value. Exact values (no grid); the other groups on the baseline |
| Runs | Plain training runs on held-out seeds 100-107 (one standard error about 0.045 pp) |

### *Held-out TTA of the oracle staircases*

| Batch size, transform | n = 5: q 0.5 / 0.75 / 1 | n = 10: q 0.5 / 0.75 / 1 | Baseline (do-nothing) |
|---|---|---|---|
| BS 2000, none | - / - / - | - / - / - | 93.92 |
| BS 2000, ramp | - / - / - | - / - / - | 93.92 (93.42) |
| BS 2000, reverse ramp | - / - / - | - / - / - | 93.92 (93.57) |
| BS 2000, wave | - / - / - | - / - / - | 93.92 (93.72) |
| BS 500, none | - / 93.85 / 93.70 | - / 94.00 / 93.88 | 94.00 |
| BS 500, ramp | - / 93.75 / 93.68 | - / 94.05 / 93.89 | 94.00 (93.37) |
| BS 500, reverse ramp | - / 93.91 / 93.62 | - / 93.88 / 93.84 | 94.00 (93.68) |
| BS 500, wave | - / 93.86 / 93.59 | - / 93.89 / 93.85 | 94.00 (93.77) |
| BS 125, none | - / 93.80 / 93.72 | - / 93.81 / 93.84 | 93.86 |
| BS 125, ramp | - / 93.72 / 93.57 | - / 93.90 / 93.84 | 93.86 (93.48) |
| BS 125, reverse ramp | - / 93.84 / 93.57 | - / 93.86 / 93.71 | 93.86 (93.68) |
| BS 125, wave | - / 93.78 / 93.55 | - / 93.95 / 93.82 | 93.86 (93.69) |

### *Observations*

Ten constant intervals can represent the baseline at BS 500 and 125; five cannot quite. With n = 10 and q = 0.75 the staircase scores 94.00 / 94.05 / 93.88 / 93.89 at BS 500 (no transform / ramp / reverse ramp / wave; baseline 94.00) and 93.81 / 93.90 / 93.86 / 93.95 at BS 125 (baseline 93.86). With n = 5 the best projection (q = 0.75) scores 93.75-93.91 at BS 500 and 93.72-93.84 at BS 125, 0.1-0.25 pp short.

The interval value should sit a bit past the interval's middle: q = 0.75 beats the end value (q = 1) in all 8 BS 500 / 125 cells with n = 5, by 0.07-0.29 pp (BS 500 reverse ramp: 93.91 vs 93.62). At BS 2000 the 5-interval schedule C4, near the end values, matched the baseline (93.95, Experiment 5).

The q = 0.5 (interval mean) and BS 2000 rows are still running.

## Experiment 17. Averaging two searches' schedules

### *Question*

Two search seeds of the same ICS version often commit different schedules, one or two grid steps apart in a few intervals, and score 0.1-0.4 pp apart. Their decision errors are partly independent; does the geometric mean of the two schedules, interval by interval, score better than the two do on average?

### *Setup*

| Item | Value |
|---|---|
| Schedules | For each finished cell of ICS v1, v2 and v5 (conv only): the geometric mean of search seeds 0 and 1's committed conv values, interval by interval (rounded to 2 significant figures, not snapped to the grid) |
| Runs | Plain training runs on held-out seeds 100-107 |

### *Held-out TTA*

| Cell | Averaged | Parents (seed 0 / 1) | Gain over their mean | Baseline |
|---|---|---|---|---|
| v1, BS 125, none | **93.77** | 93.70 / 93.76 | +0.04 | 93.86 |
| v1, BS 125, reverse ramp | **93.76** | 93.72 / 93.60 | +0.10 | 93.86 |
| v1, BS 2000, none | **93.90** | 93.82 / 93.82 | +0.08 | 93.92 |
| v1, BS 2000, ramp | **93.16** | 93.19 / 93.11 | +0.02 | 93.92 |
| v1, BS 2000, reverse ramp | **93.89** | 93.81 / 93.79 | +0.10 | 93.92 |
| v1, BS 2000, wave | **93.63** | 93.61 / 93.46 | +0.09 | 93.92 |
| v1, BS 500, none | **93.81** | 93.75 / 93.92 | -0.03 | 94.00 |
| v1, BS 500, ramp | **93.43** | 93.27 / 93.53 | +0.04 | 94.00 |
| v1, BS 500, reverse ramp | **93.95** | 93.95 / 93.82 | +0.07 | 94.00 |
| v1, BS 500, wave | **93.66** | 93.25 / 93.75 | +0.16 | 94.00 |
| v2, BS 125, none | **93.66** | 93.43 / 93.52 | +0.18 | 93.86 |
| v2, BS 125, ramp | **93.49** | 93.50 / 93.25 | +0.12 | 93.86 |
| v2, BS 125, reverse ramp | **93.74** | 93.72 / 93.78 | -0.01 | 93.86 |
| v2, BS 125, wave | **93.70** | 93.40 / 93.70 | +0.15 | 93.86 |
| v2, BS 2000, none | **93.81** | 93.77 / 93.62 | +0.12 | 93.92 |
| v2, BS 2000, ramp | **93.74** | 93.68 / 93.77 | +0.02 | 93.92 |
| v2, BS 2000, reverse ramp | **93.78** | 93.83 / 93.71 | +0.01 | 93.92 |
| v2, BS 2000, wave | **93.78** | 93.84 / 93.53 | +0.10 | 93.92 |
| v2, BS 500, none | **93.90** | 93.89 / 93.88 | +0.02 | 94.00 |
| v2, BS 500, ramp | **93.66** | 93.54 / 93.68 | +0.05 | 94.00 |
| v2, BS 500, reverse ramp | **93.92** | 93.81 / 93.81 | +0.11 | 94.00 |
| v2, BS 500, wave | **93.82** | 93.70 / 93.85 | +0.04 | 94.00 |
| v5, BS 2000, none | **93.75** | 93.95 / 93.56 | -0.01 | 93.92 |
| v5, BS 2000, ramp | **93.62** | 93.66 / 93.51 | +0.04 | 93.92 |
| v5, BS 2000, reverse ramp | **93.86** | 93.91 / 93.71 | +0.06 | 93.92 |
| v5, BS 2000, wave | **93.85** | 93.84 / 93.75 | +0.06 | 93.92 |

Mean gain over the parents' mean: +0.066 pp over 26 cells (23 positive).

### *Observations*

Averaging two search seeds' schedules helps on average: the averaged schedule beats the mean of its two parents by +0.066 pp over 26 cells (v1, v2 and v5, conv only), positive in 23 of them, by up to +0.18 pp (v2 at BS 125). It mostly recovers the better parent: when one seed made a costly mistake, the average sits between the two choices, and the score curves are flat enough near the optimum that the midpoint costs little.

This is decision noise being averaged out: the search's single-seed decisions err by a grid step in some intervals (Experiment 12), and two independent searches err in different intervals. Two v1 searches averaged (about 20 full runs together) reach 93.90 at BS 2000 without a transform and 93.89 under the reverse ramp, against 93.75 and

93.81 for one v5 search (about 15 full runs); averaging cannot fix a shared bias, though (v1 under the ramp stays at 93.16).

## Experiment 18. Expanding 5-interval schedules to 10 intervals

### *Question*

Ten intervals represent the baseline at BS 500 and 125 where five fall short (Experiment 16), but a 10-interval search makes twice the noisy decisions. Does a 5-interval schedule, expanded to 10 intervals by a fixed rule, score like a 10-interval one?

### *Setup*

| Item | Value |
|---|---|
| Expansion | Each interval's value v is split into two halves: v x (previous / v)^(1/4) and v x (next / v)^(1/4) (the first interval's first half stays v; the last interval's second half continues the decay, v x (v / previous)^(1/4)). A quarter of the way, in log, toward each neighbour: the staircase becomes twice as fine and follows the trend |
| Schedules | 5-interval oracle staircases (q = 0.75, Experiment 16), searched schedules (ICS v2, v5) and averaged ones (Experiment 17); held-out seeds 100-107 |

### *Held-out TTA*

| 5-interval schedule | As 5 intervals | Expanded to 10 | Gain | Baseline |
|---|---|---|---|---|
| averaged v1, BS 2000, none | 93.90 | **93.88** | -0.02 | 93.92 |
| averaged v2, BS 125, none | 93.66 | **93.68** | +0.02 | 93.86 |
| averaged v2, BS 500, none | 93.90 | **93.99** | +0.08 | 94.00 |
| averaged v2, BS 500, wave | 93.82 | **93.79** | -0.02 | 94.00 |
| averaged v5, BS 2000, none | 93.75 | **93.78** | +0.03 | 93.92 |
| oracle staircase q 0.75, BS 125, none | 93.80 | **93.95** | +0.14 | 93.86 |
| oracle staircase q 0.75, BS 125, wave | 93.78 | **93.96** | +0.18 | 93.86 |
| oracle staircase q 0.75, BS 500, none | 93.85 | **94.03** | +0.18 | 94.00 |
| oracle staircase q 0.75, BS 500, ramp | 93.75 | **94.00** | +0.25 | 94.00 |
| v2 seed 0, BS 2000, none | 93.77 | **93.83** | +0.06 | 93.92 |
| v2 seed 1, BS 2000, none | 93.62 | **93.83** | +0.21 | 93.92 |
| v2 seed 0, BS 500, none | 93.89 | **93.87** | -0.02 | 94.00 |
| v2 seed 1, BS 500, none | 93.88 | **93.91** | +0.03 | 94.00 |
| v5 seed 0, BS 2000, none | 93.95 | **93.98** | +0.03 | 93.92 |
| v5 seed 1, BS 2000, none | 93.56 | **93.61** | +0.05 | 93.92 |
| v5 seed 0, BS 500, none | 93.83 | **93.90** | +0.07 | 94.00 |
| v5 seed 1, BS 500, none | 93.89 | **94.00** | +0.11 | 94.00 |

Mean gain +0.082 pp over 17 schedules (14 positive).

### *Observations*

Expansion closes the representation gap: the 5-interval oracle staircases gain 0.14-0.25 pp and reach the baseline (BS 500: 93.85 -> 94.03, ramp 93.75 -> 94.00; BS 125: 93.80 -> 93.95, wave 93.78 -> 93.96), as high as the 10-interval staircases of Experiment 16.

Searched schedules gain less but rarely lose: v5's BS 500 seed 1 goes from 93.89 to 94.00 and seed 0 from 93.83 to 93.90, v2's BS 2000 seed 1 from 93.62 to 93.83; the few losses are 0.02 pp. A searched schedule's errors are not the staircase's coarseness, which expansion fixes, but its decisions, which it does not.

So the search can stay at 5 intervals (half the decisions of a 10-interval search, so less decision noise and lower cost: 10-interval searches tried in development, ICS v7 and v8, scored 93.64-93.83 on 2 held-out seeds) and expand its result for free.

## Experiment 19. ICS v9-v11: 5-interval search expanded to 10 intervals, with a scaled or fixed plan tail

### *Question*

Searching 5 intervals and expanding the result to 10 (Experiment 18), which second-pass lookahead works: the previous pass's tail scaled by candidate / plan value (v9), or left as it is so a candidate changes only its own interval (v10, v11)? And should the budget go to replicated decisions (v10) or wider climbs (v11)?

### *Setup*

| Item | Value |
|---|---|
| Common | 5 constant intervals; pass 1: climb each interval from the previous one's value with a one-interval cooldown at 0.22 x the candidate; pass 2: from the initial state, keep interval 1, climb intervals 2-5 from pass 1's values with a rest-of-run lookahead on pass 1's schedule; the result expanded to 10 intervals (Experiment 18). 4 LRs from 1.0: a joint x7.7 level scan of head and the biases (largest within 1 pp after a one-interval cooldown), then each alone at most two x7.7 steps up; the other groups follow conv's profile. Budget 20 full runs; 2 search seeds; held-out seeds 100-107 |
| v9 | Pass-2 tail scaled by candidate / pass-1 value; climbs at most 1 grid step from pass 1; decisions replicated over 2 continuations (3 in the last interval; 4 LRs: only intervals 4-5) |
| v10 | As v9 with the tail unscaled (rest_fixed) |
| v11 | As v10 with replicates only in intervals 4-5 and climbs of up to 2 grid steps |

### *Held-out TTA (mean over search seeds; per seed); cost in full runs*

| Cell (baseline; do-nothing) | v9 | v10 | v11 |
|---|---|---|---|
| BS 2000, conv only, none (93.92) | **93.79** (93.80/93.79); 18x | **93.89** (93.86/93.92); 17x | **93.94** (93.97/93.91); 15x |
| BS 2000, conv only, ramp (93.92; 93.42) | - | **93.82** (93.82/93.82); 20x* | **93.80** (93.75/93.85); 19x |
| BS 2000, conv only, reverse ramp (93.92; 93.57) | - | **93.78** (93.69/93.88); 19x | **93.88** (93.89/93.88); 17x |
| BS 2000, conv only, wave (93.92; 93.72) | - | **93.88** (93.89/93.88); 19x | **93.88** (93.89/93.88); 16x |
| BS 2000, 4 LRs, none (93.92) | - | **93.40** (93.75/93.06); 19x | **93.39** (93.74/93.04); 20x* |
| BS 2000, 4 LRs, ramp (93.92; 93.25) | - | **93.26** (93.42/93.10); 20x* | **93.42** (93.42); 21x* |
| BS 2000, 4 LRs, reverse ramp (93.92; 93.45) | - | **93.75** (93.81/93.70); 17x | - |
| BS 2000, 4 LRs, wave (93.92; 93.68) | - | **93.62** (93.64/93.60); 20x | - |
| BS 500, conv only, none (94.00) | - | **94.00** (94.00/94.00); 19x | - |
| BS 500, conv only, ramp (94.00; 93.37) | - | **93.72** (93.66/93.77); 20x* | - |
| BS 500, conv only, reverse ramp (94.00; 93.68) | - | **93.85** (93.88/93.82); 19x | - |
| BS 500, conv only, wave (94.00; 93.77) | - | **93.87** (93.85/93.89); 19x | - |
| BS 500, 4 LRs, none (94.00) | - | **93.35** (93.47/93.22); 20x | - |
| BS 500, 4 LRs, ramp (94.00; 93.32) | - | **92.92** (92.86/92.98); 20x* | - |
| BS 500, 4 LRs, reverse ramp (94.00; 93.60) | - | **93.66** (93.42/93.90); 19x | - |
| BS 500, 4 LRs, wave (94.00; 93.83) | - | **93.66** (93.68/93.65); 20x | - |
| BS 125, conv only, none (93.86) | - | **93.72** (93.72/93.71); 20x | - |
| BS 125, conv only, ramp (93.86; 93.48) | - | **93.70** (93.76/93.63); 20x | - |
| BS 125, conv only, reverse ramp (93.86; 93.68) | - | **93.77** (93.69/93.84); 18x | - |
| BS 125, conv only, wave (93.86; 93.69) | - | **93.76** (93.69/93.82); 19x | - |
| BS 125, 4 LRs, none (93.86) | - | **93.30** (93.44/93.17); 20x | - |
| BS 125, 4 LRs, ramp (93.86; 93.26) | - | **93.26** (93.49/93.03); 20x* | - |
| BS 125, 4 LRs, reverse ramp (93.86; 93.68) | - | **93.16** (92.73/93.60); 18x | - |
| BS 125, 4 LRs, wave (93.86; 93.70) | - | **93.67** (93.48/93.86); 19x | - |

Within 0.1 pp of the baseline or above (finished cells): v9: 0 of 1; v10: 5 of 24; v11: 3 of 6. * = the search hit its step budget.